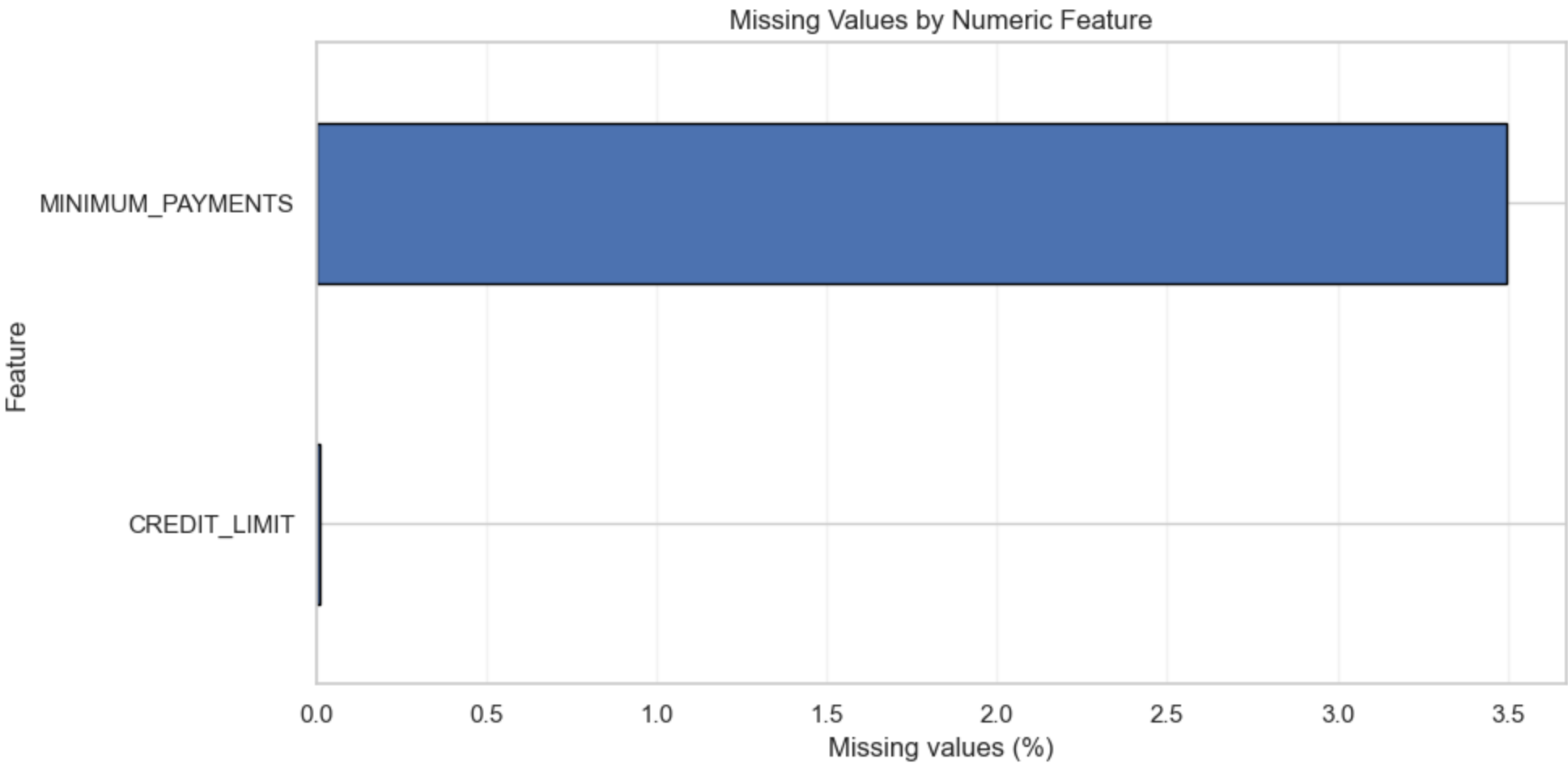


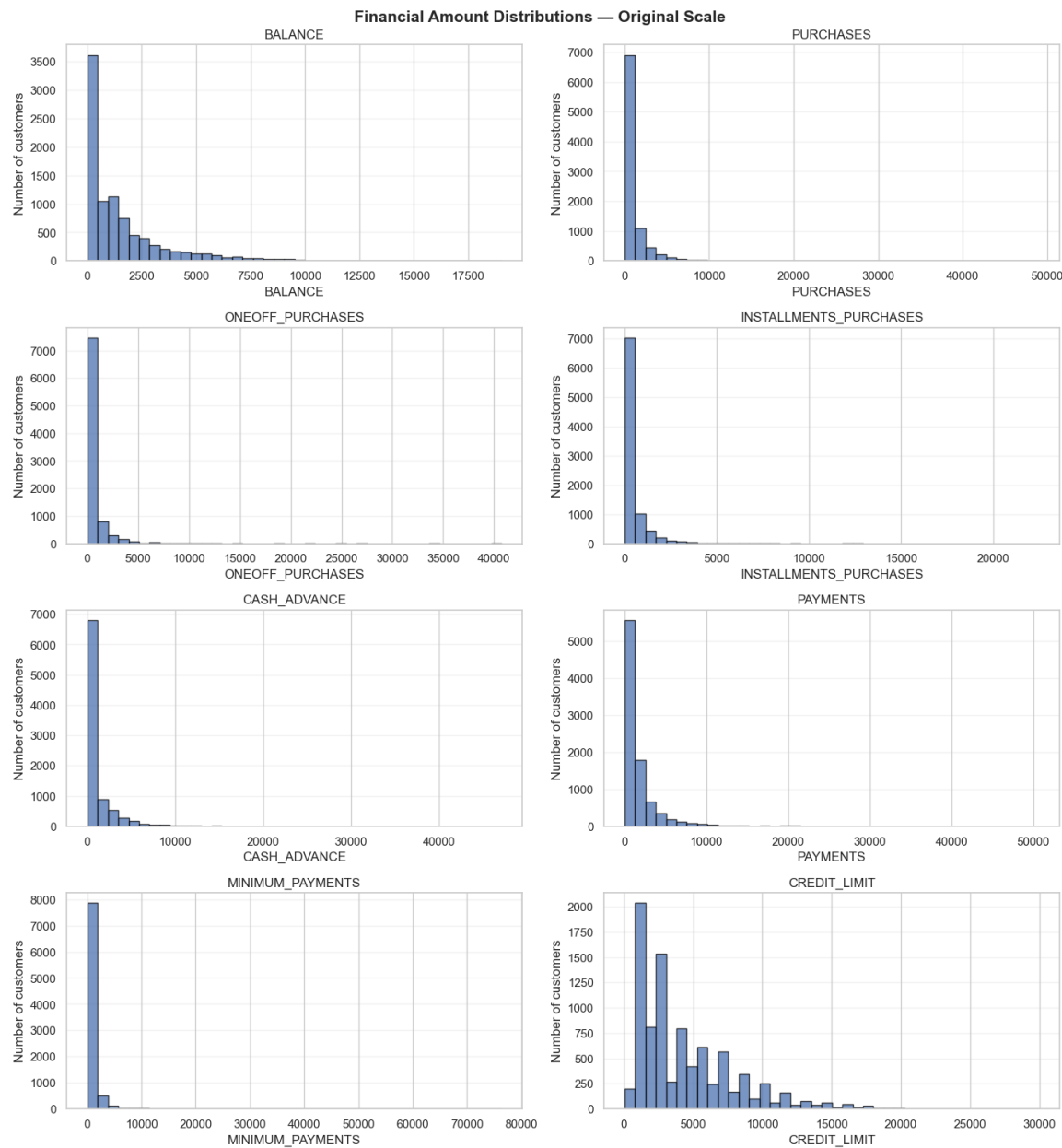
Unsupervised Exam Project

Code-Generated Graphs and Tables Only
Graphs: 25 | Tables: 83

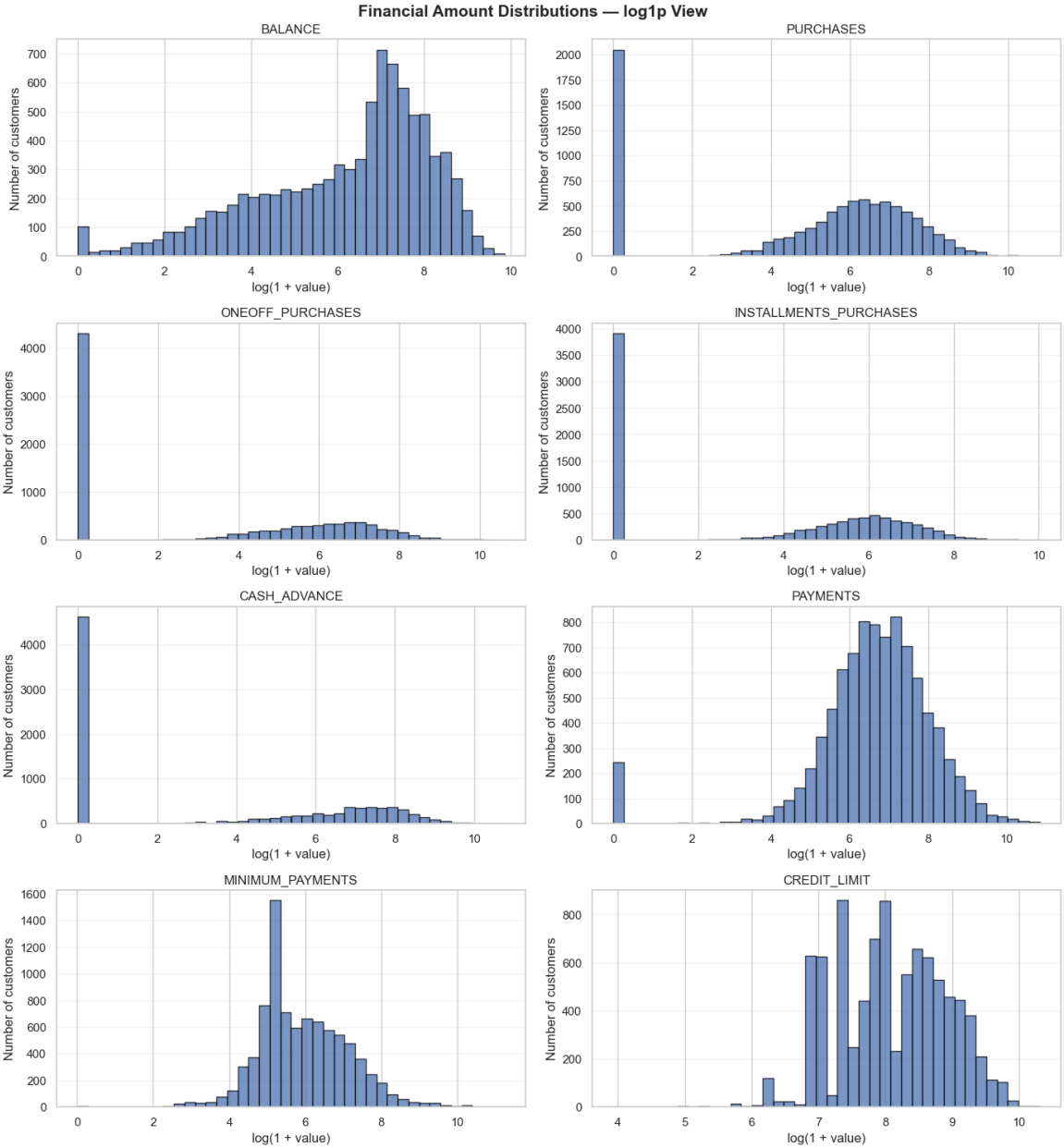
Graph 1 — Generated by Notebook Cell 3



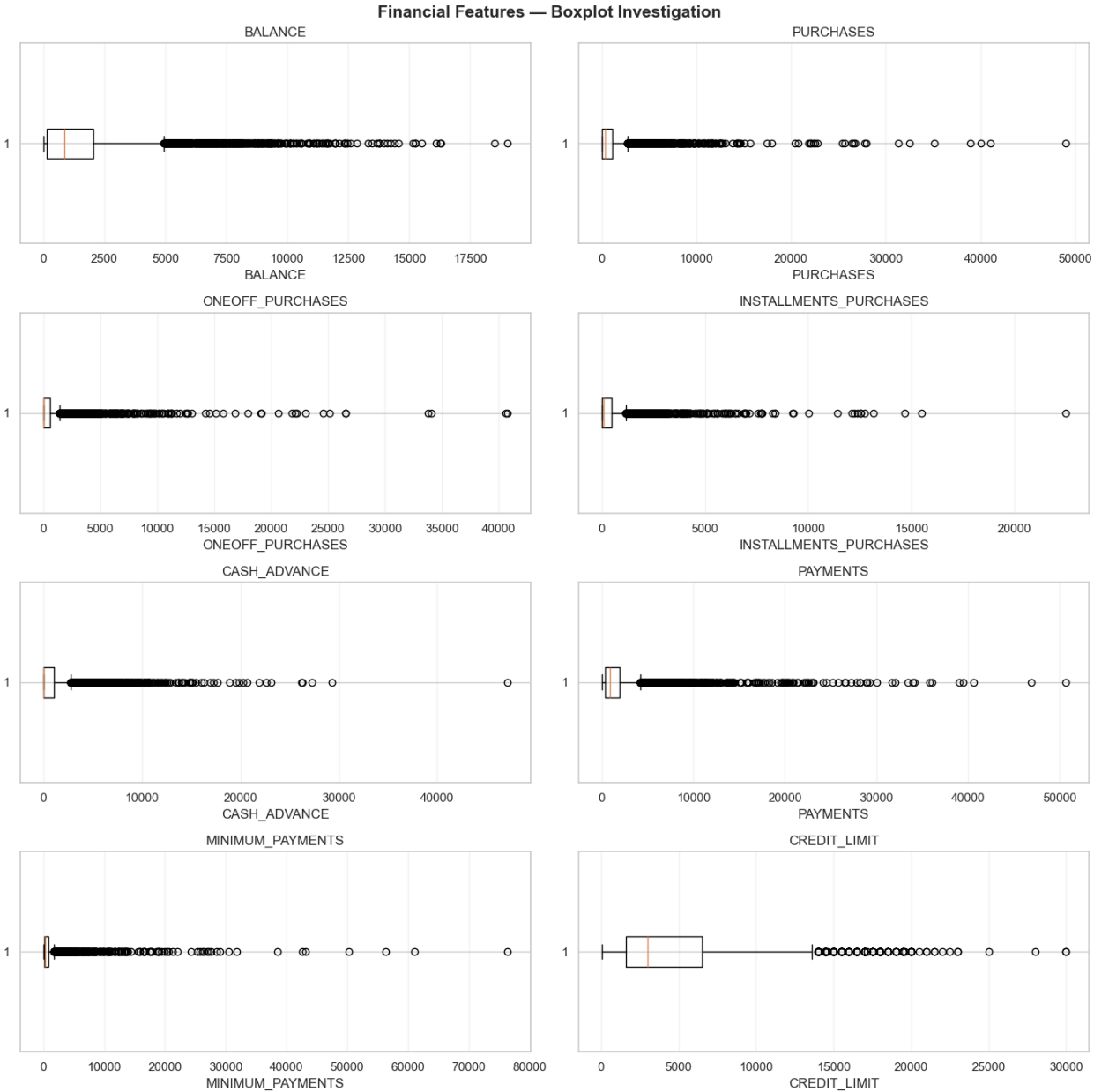
Graph 2 — Generated by Notebook Cell 3



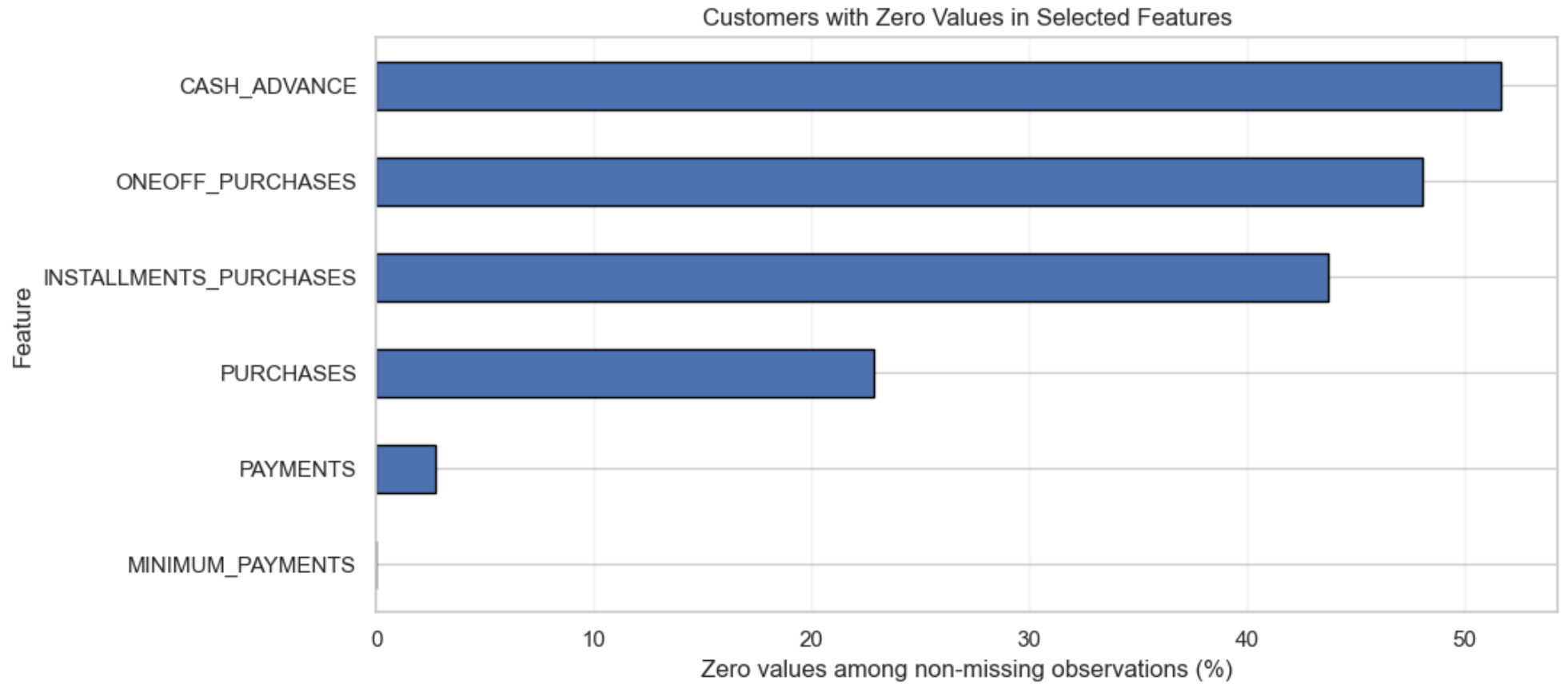
Graph 3 — Generated by Notebook Cell 3



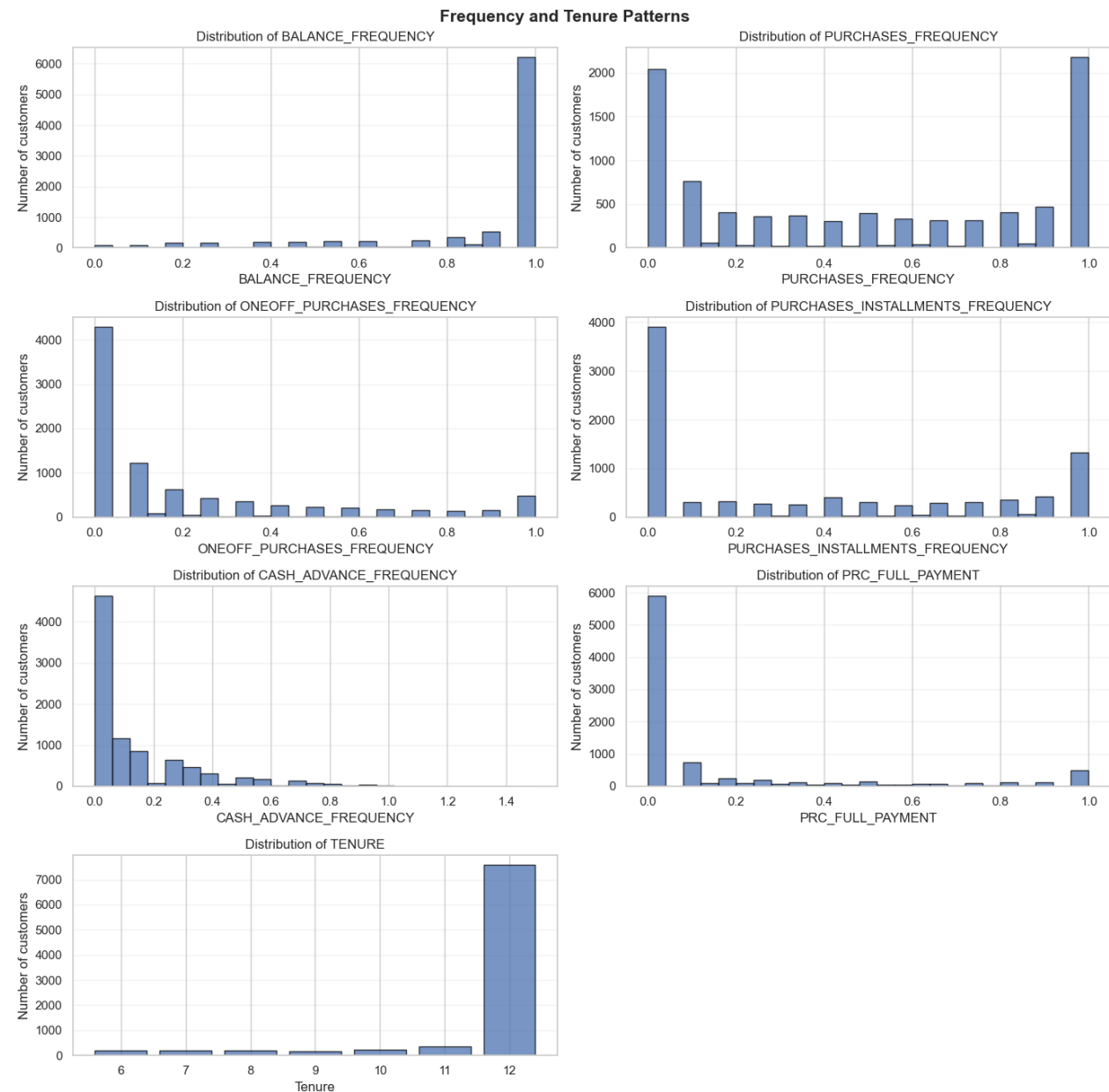
Graph 4 — Generated by Notebook Cell 3



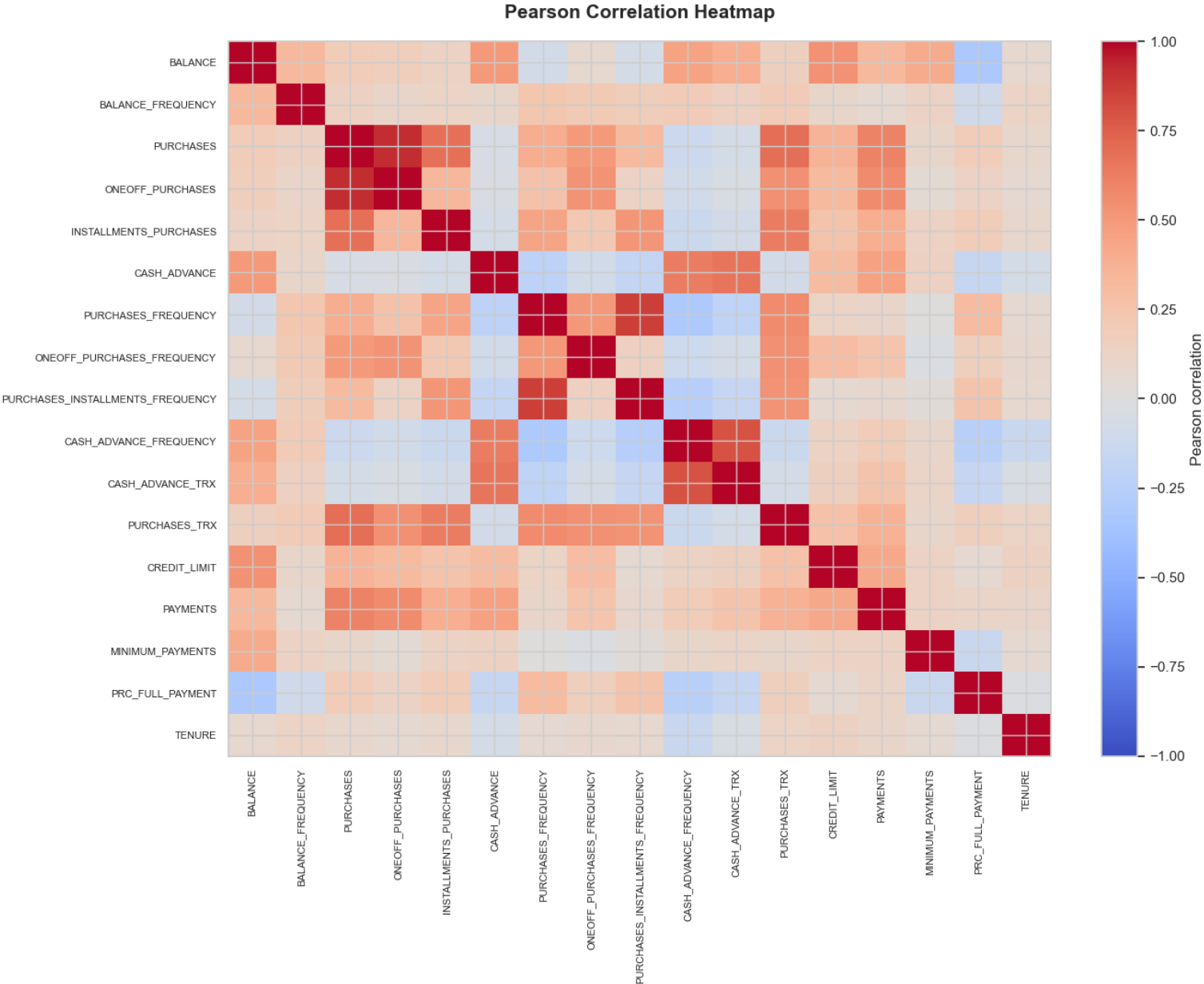
Graph 5 — Generated by Notebook Cell 3



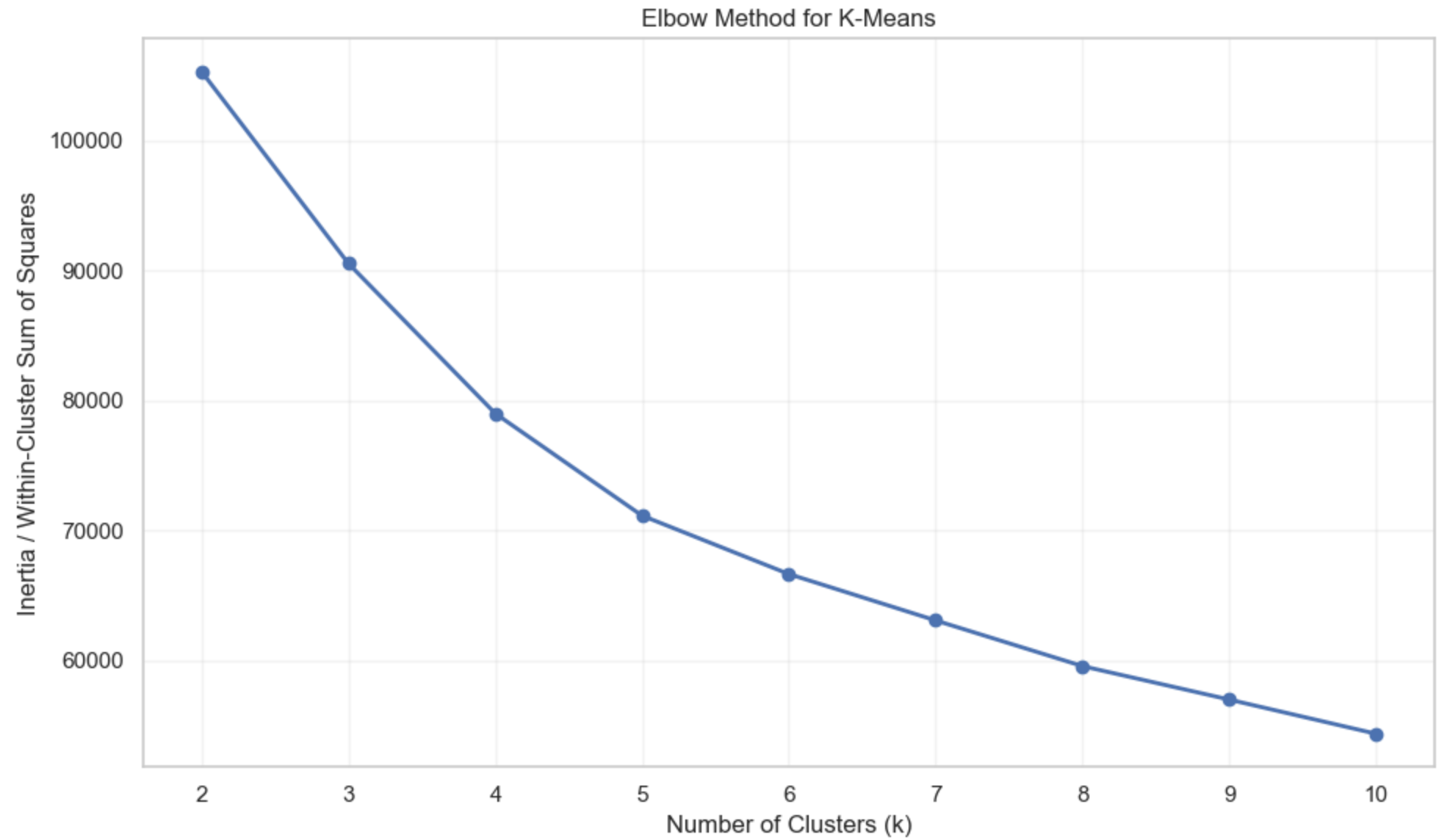
Graph 6 — Generated by Notebook Cell 3



Graph 7 — Generated by Notebook Cell 3



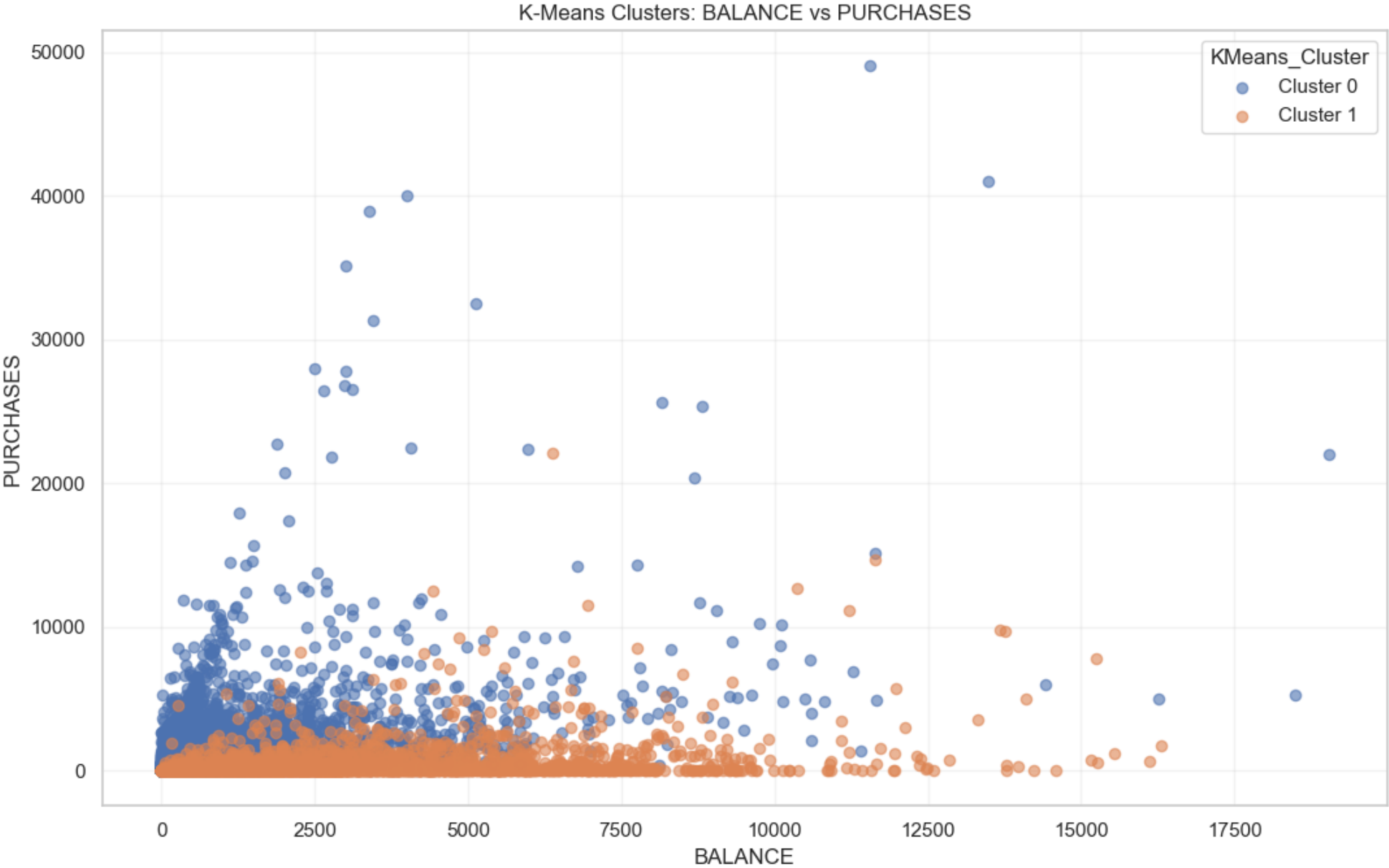
Graph 8 — Generated by Notebook Cell 5



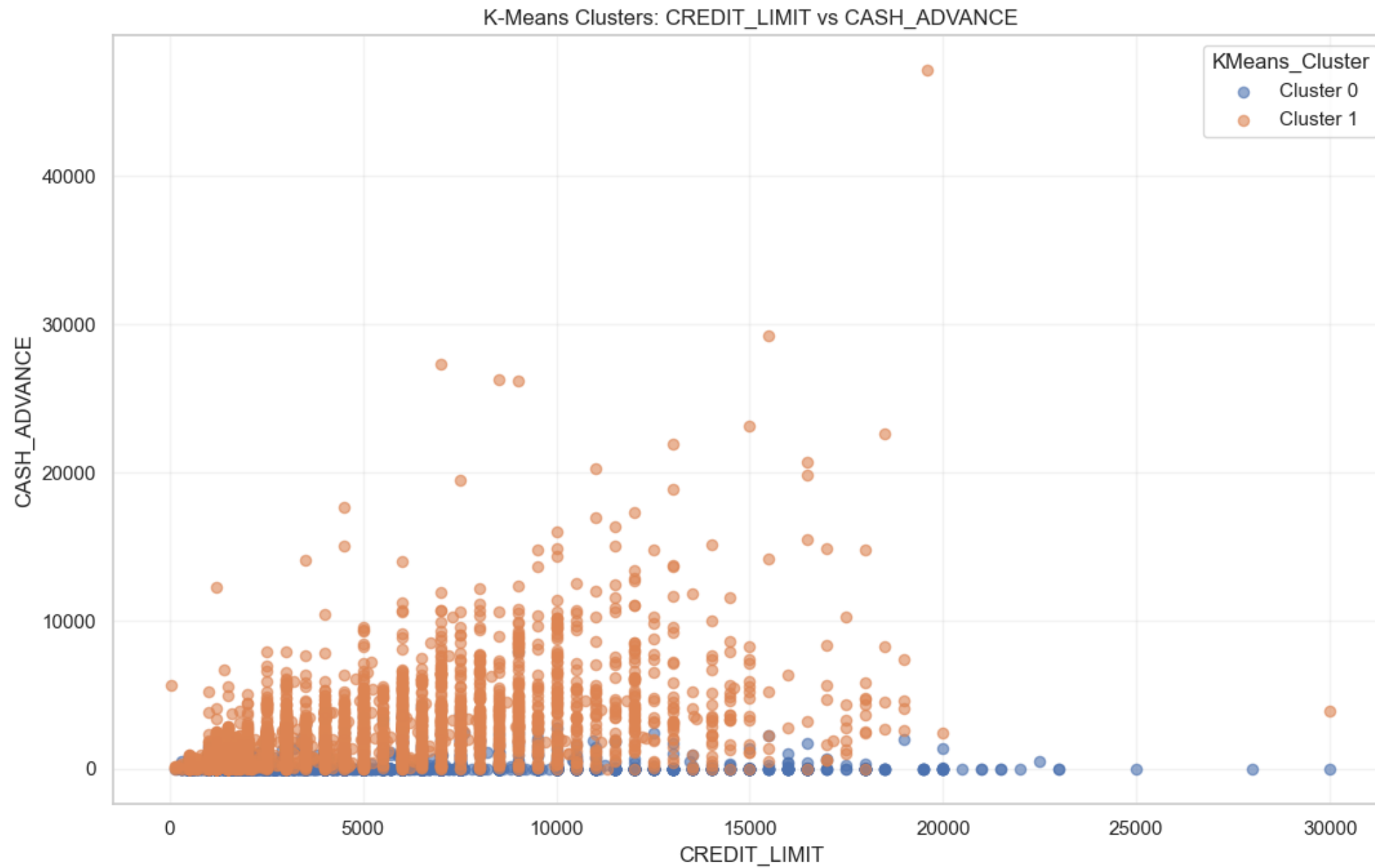
Graph 9 — Generated by Notebook Cell 5



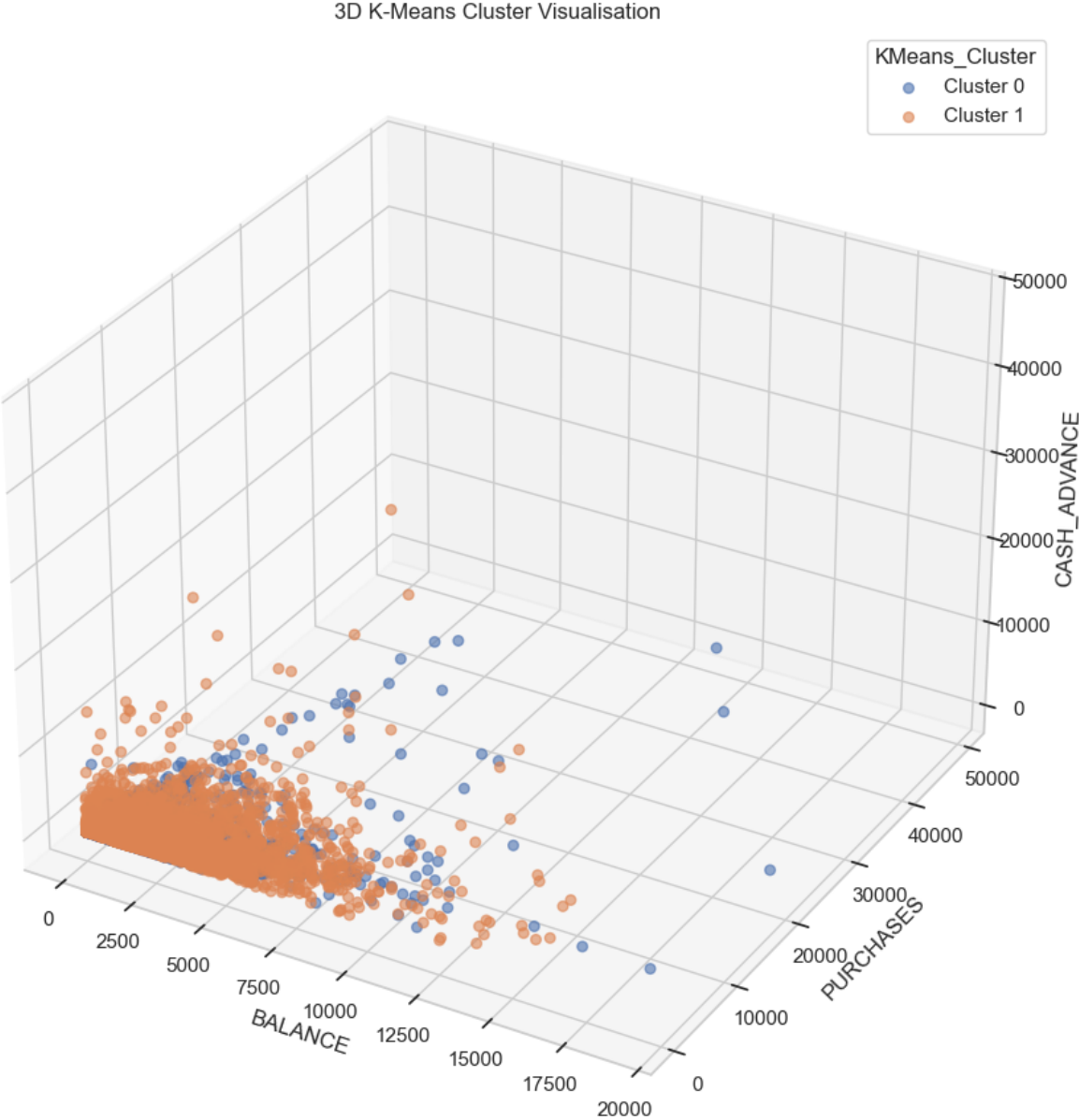
Graph 10 — Generated by Notebook Cell 5



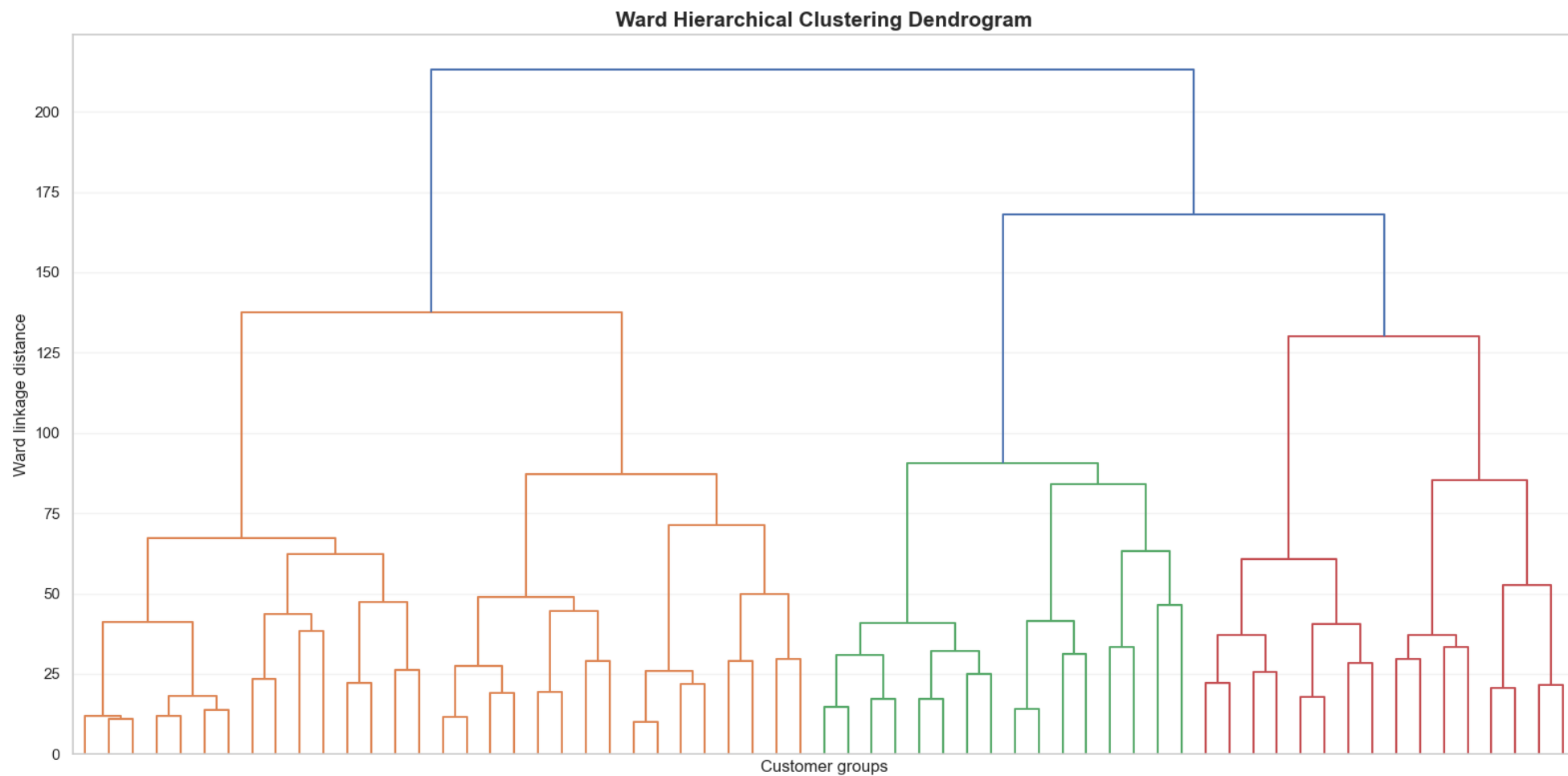
Graph 11 — Generated by Notebook Cell 5



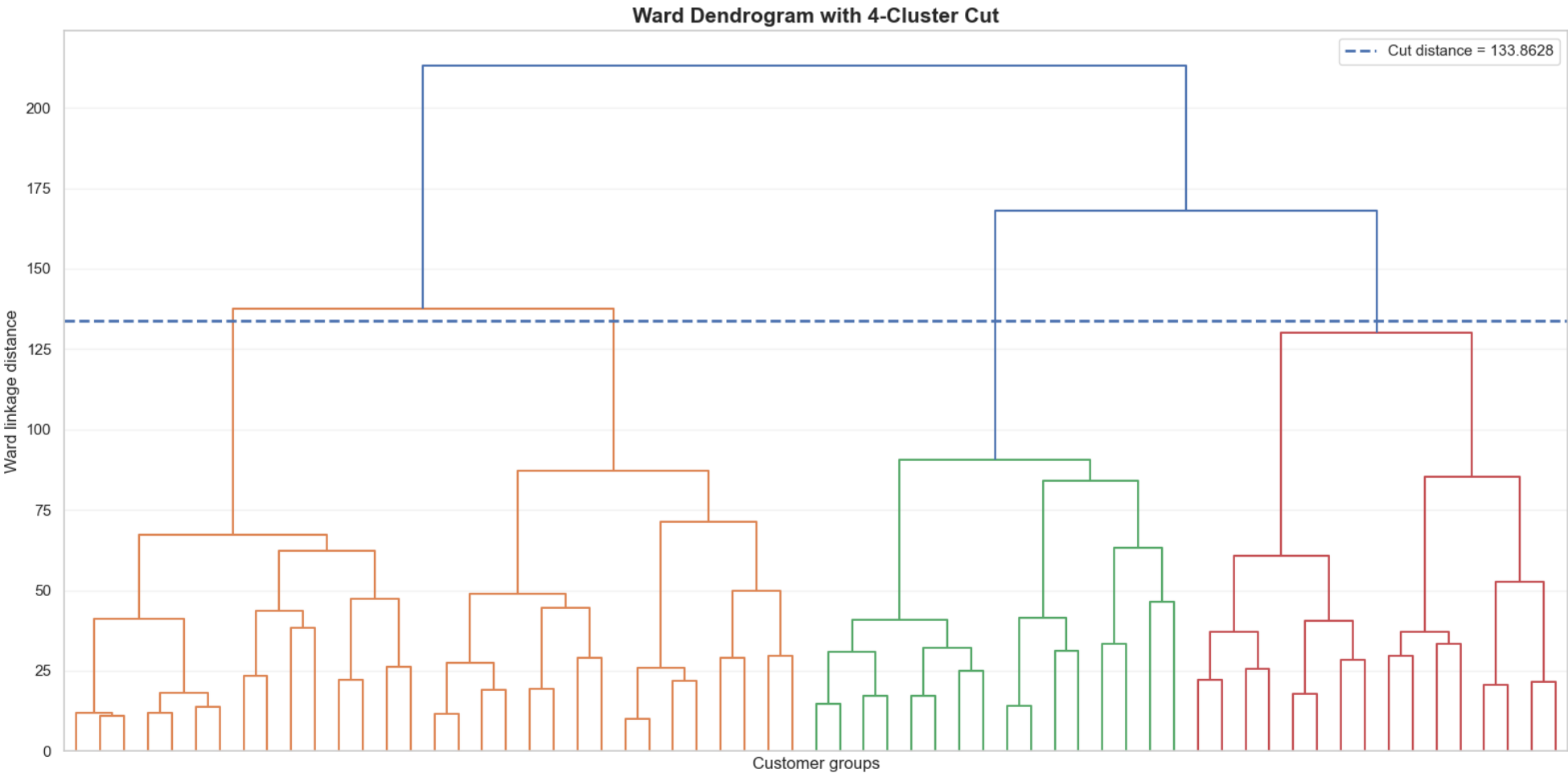
Graph 12 — Generated by Notebook Cell 5



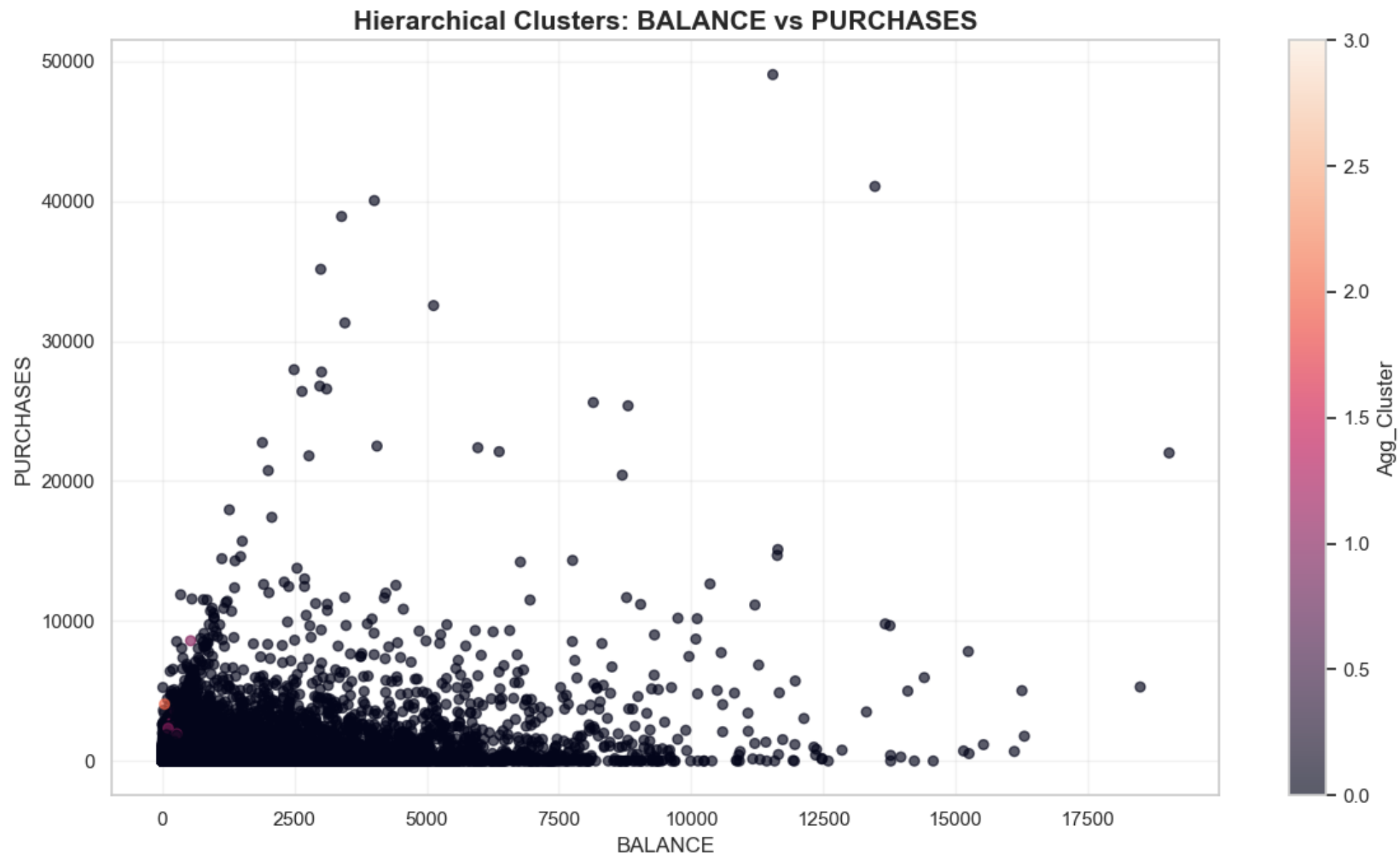
Graph 13 — Generated by Notebook Cell 6



Graph 14 — Generated by Notebook Cell 6

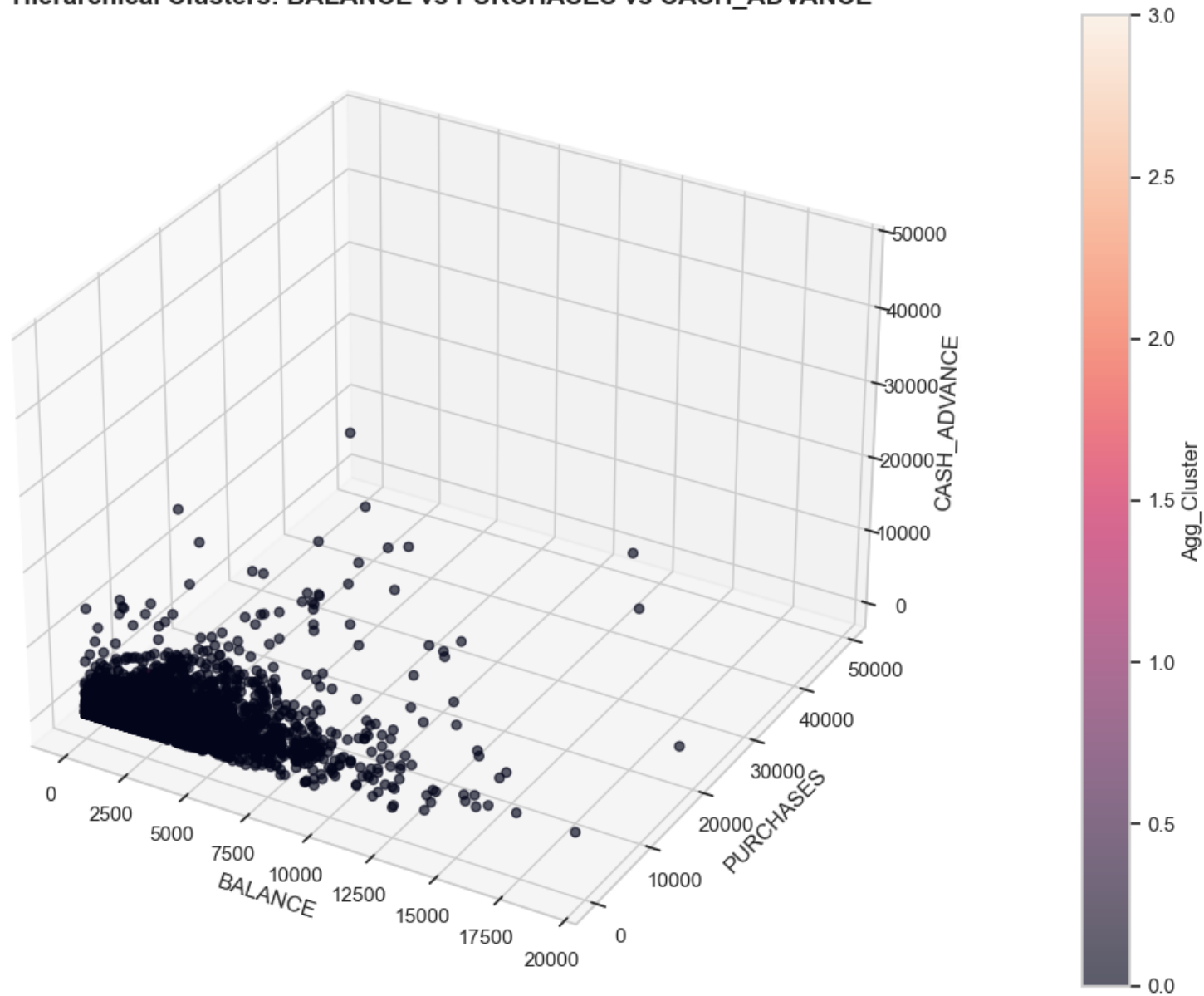


Graph 15 — Generated by Notebook Cell 6

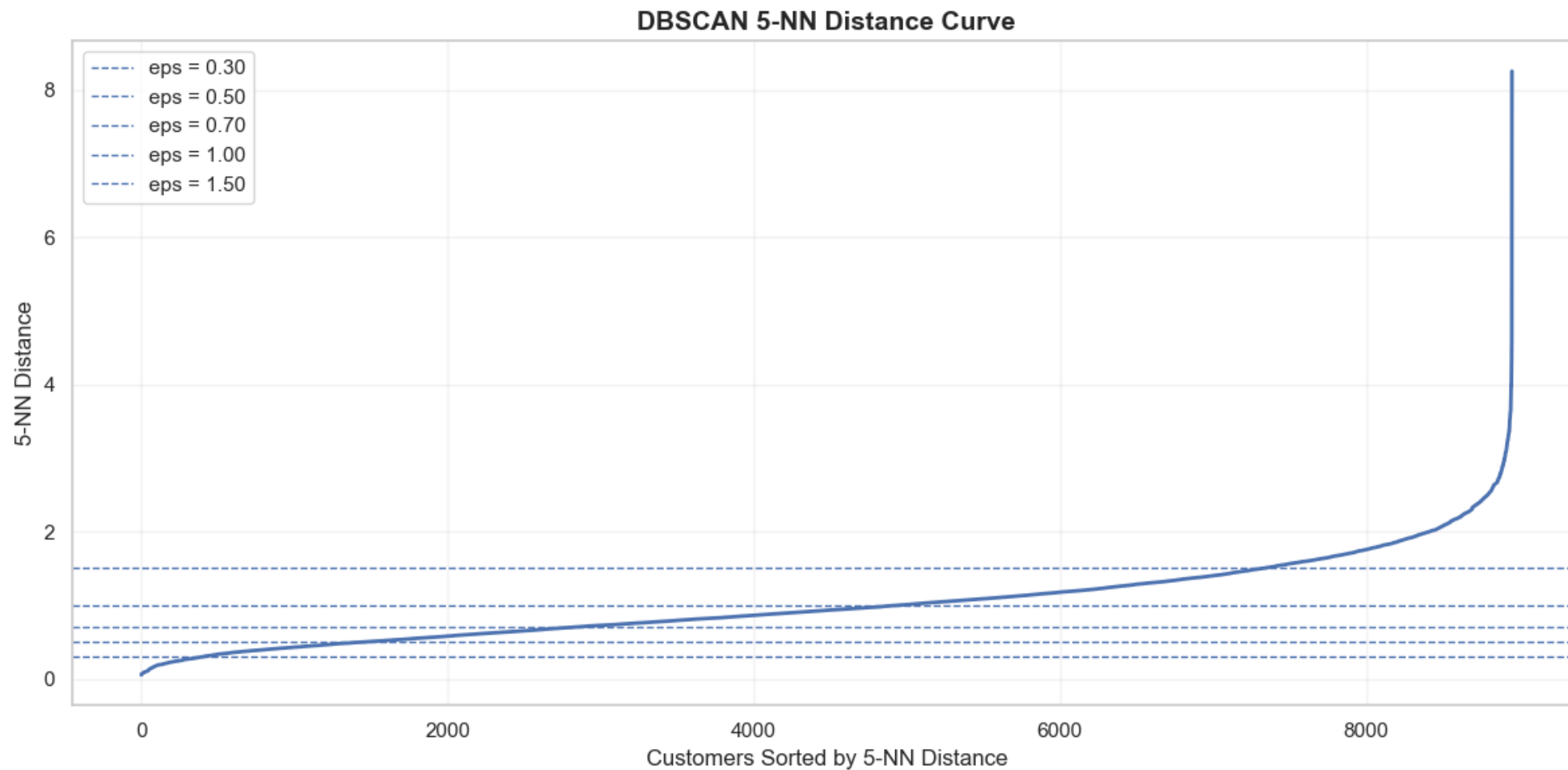


Graph 16 — Generated by Notebook Cell 6

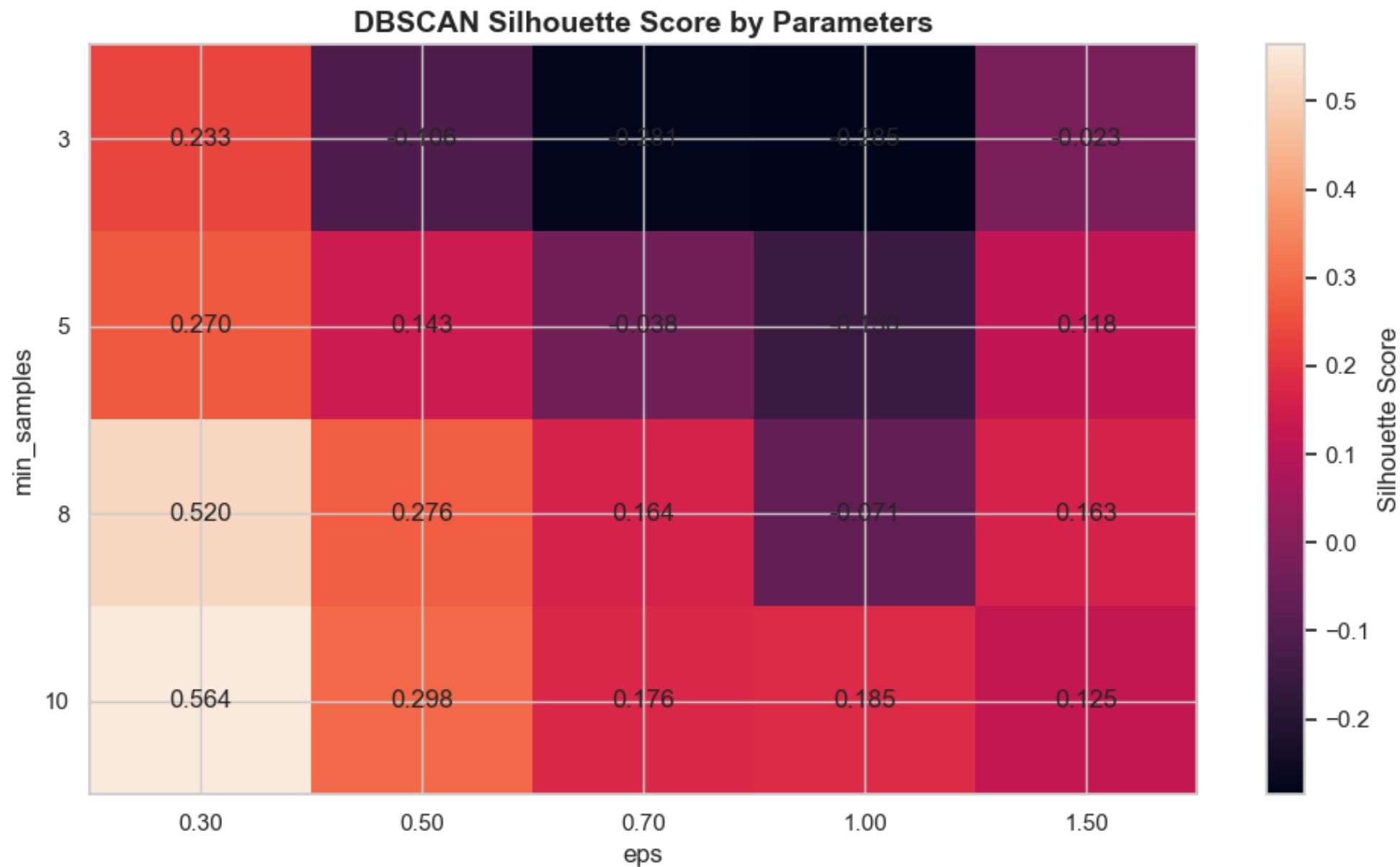
Hierarchical Clusters: BALANCE vs PURCHASES vs CASH_ADVANCE



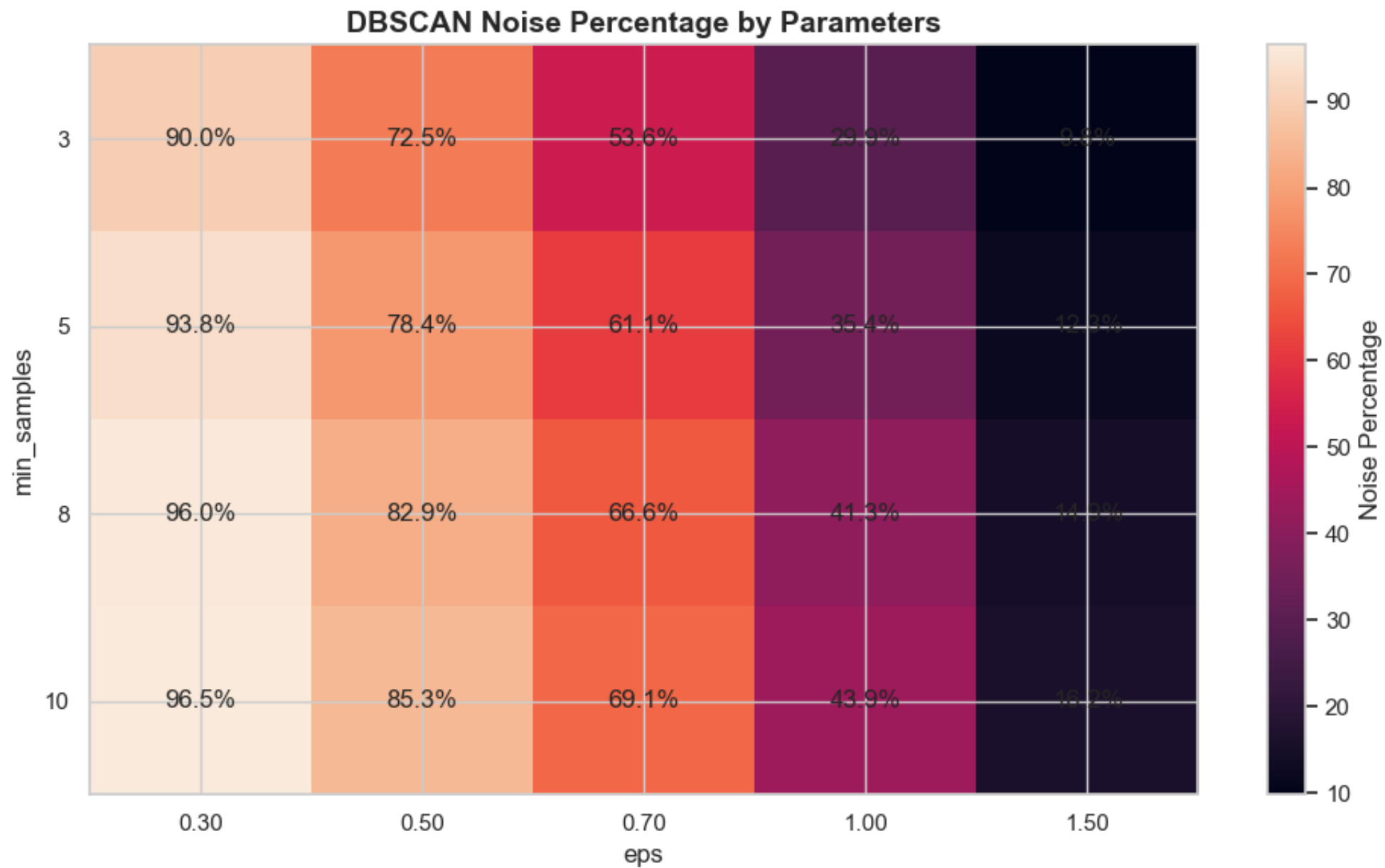
Graph 17 — Generated by Notebook Cell 7



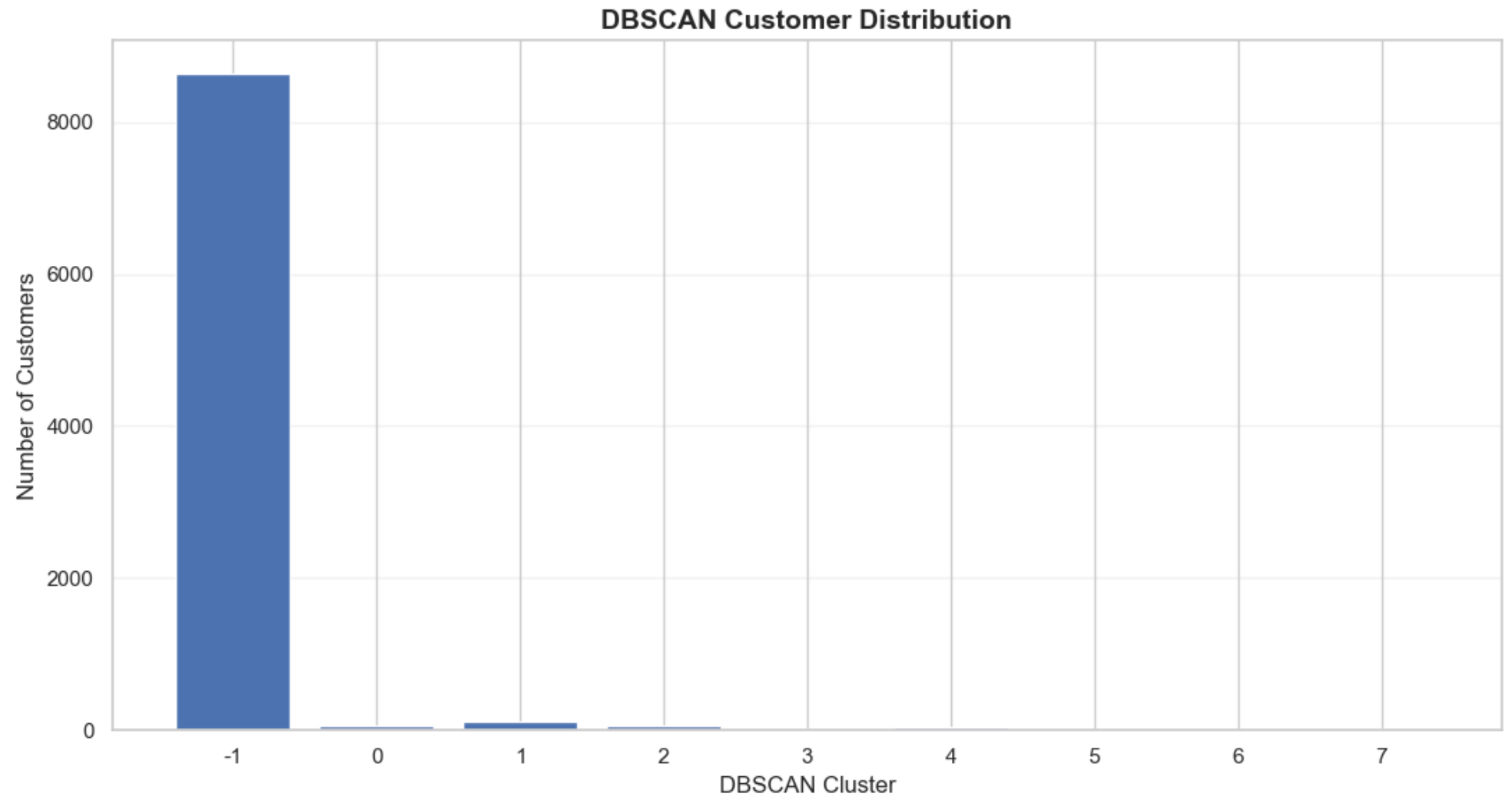
Graph 18 — Generated by Notebook Cell 7



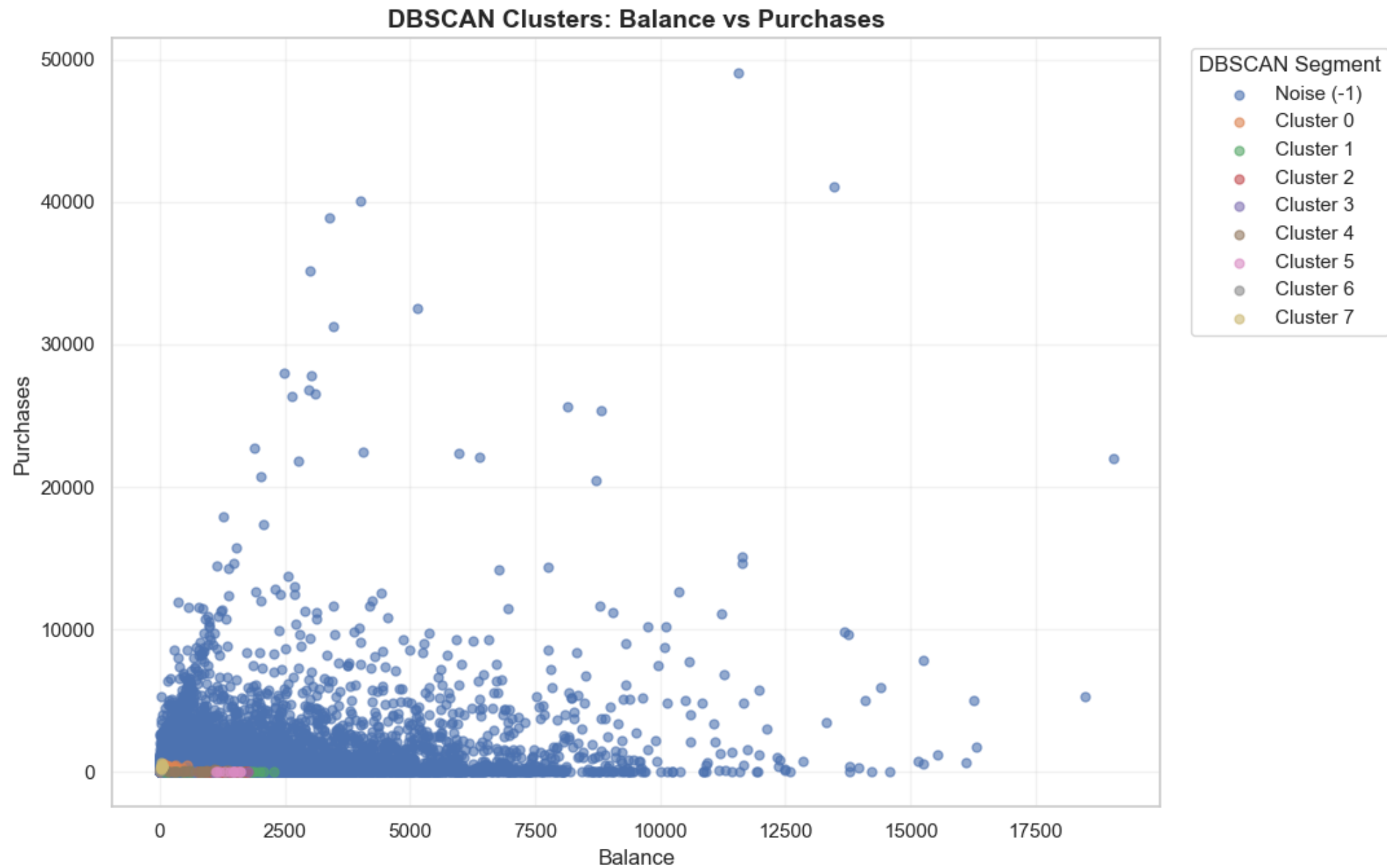
Graph 19 — Generated by Notebook Cell 7



Graph 20 — Generated by Notebook Cell 7

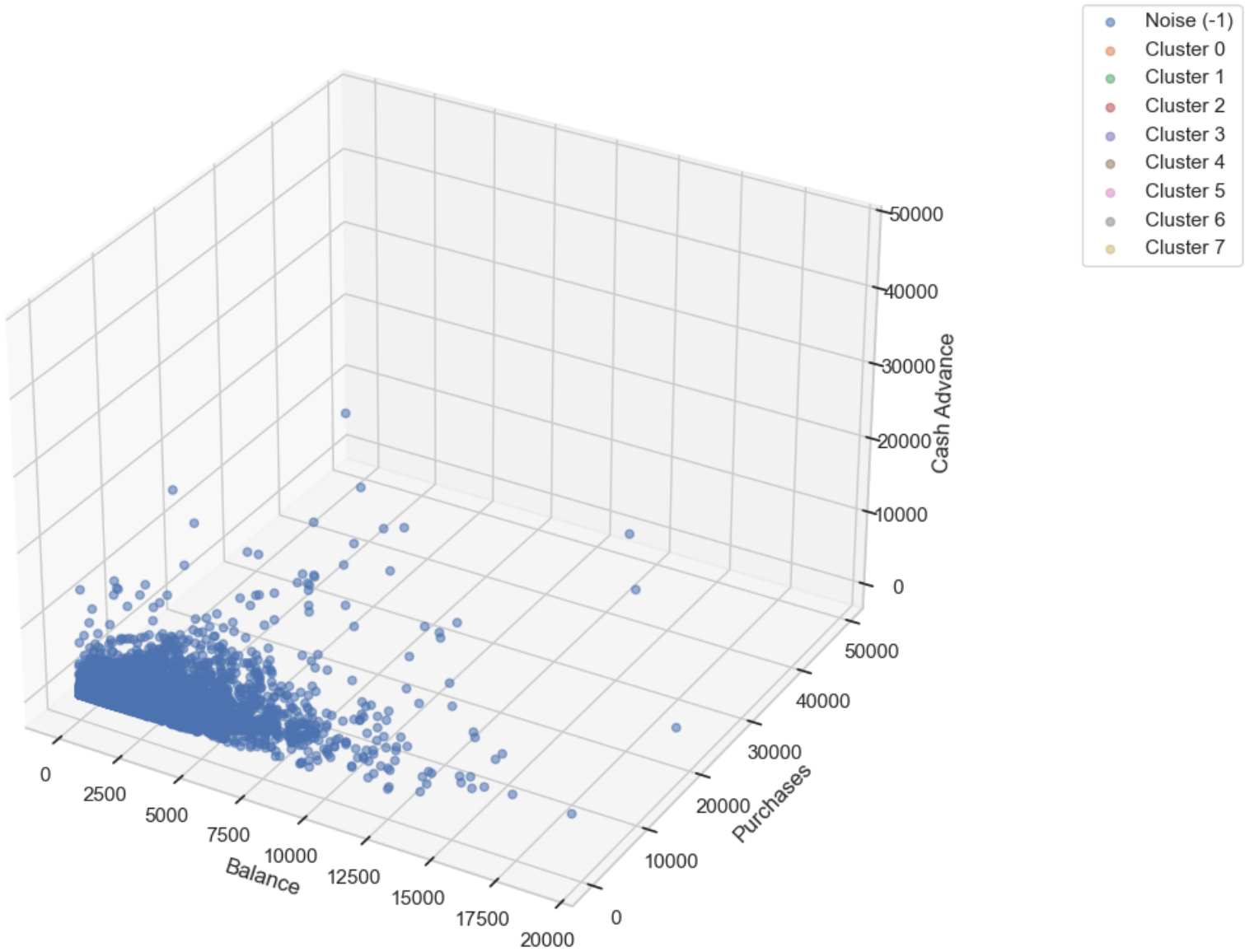


Graph 21 — Generated by Notebook Cell 7

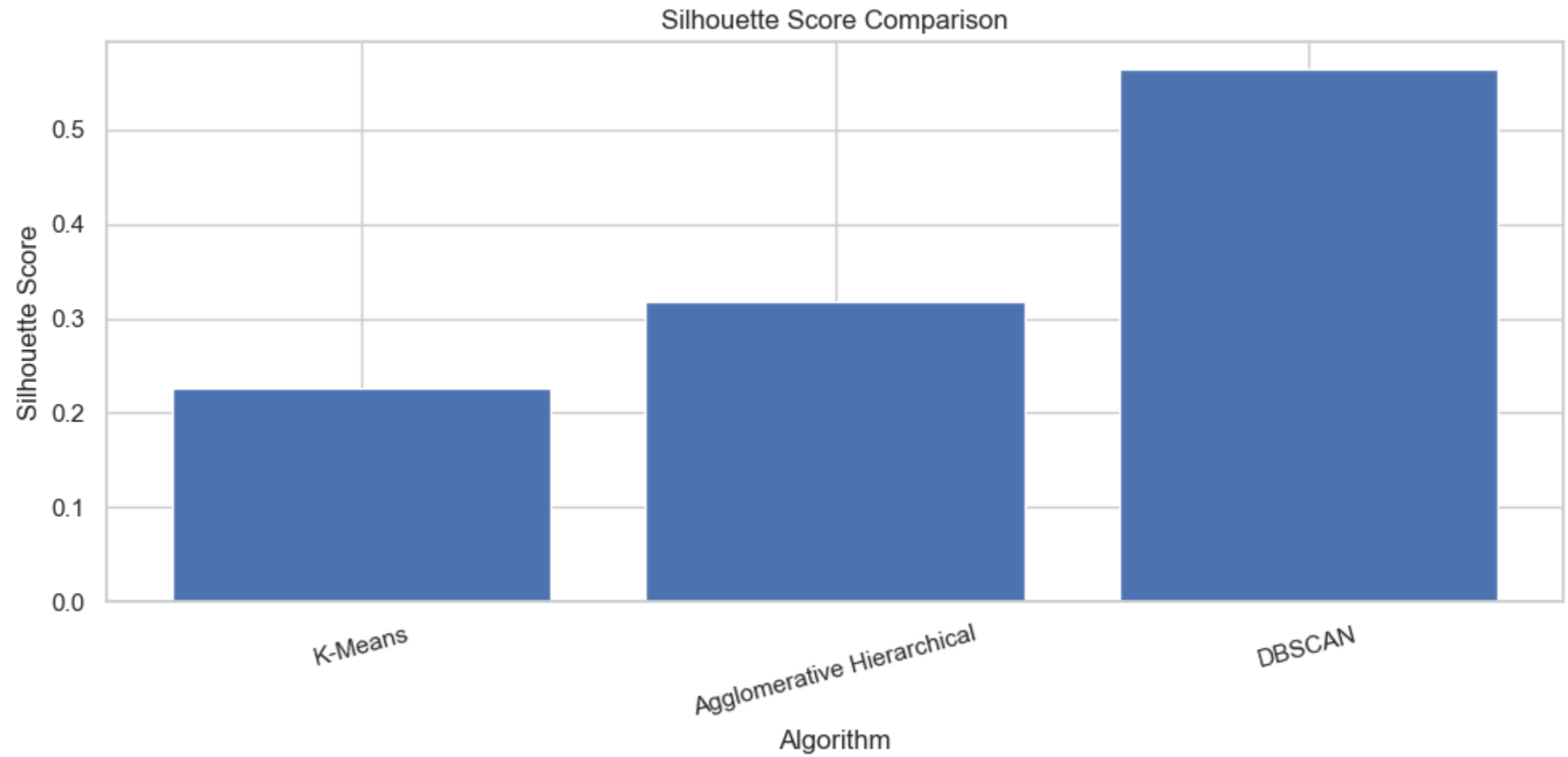


Graph 22 — Generated by Notebook Cell 7

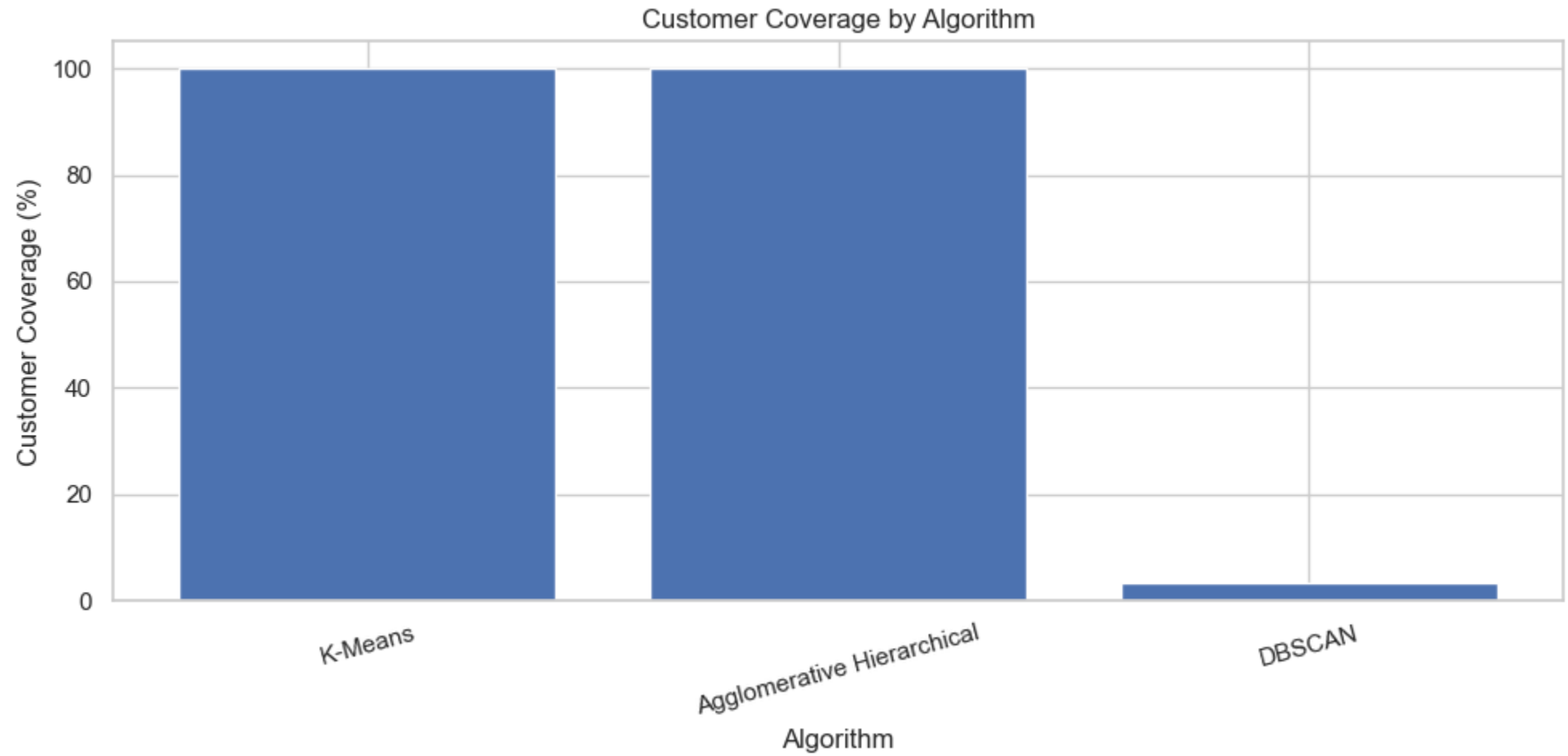
DBSCAN Clusters: Balance, Purchases & Cash Advance



Graph 23 — Generated by Notebook Cell 8



Graph 24 — Generated by Notebook Cell 8



Graph 25 — Generated by Notebook Cell 8

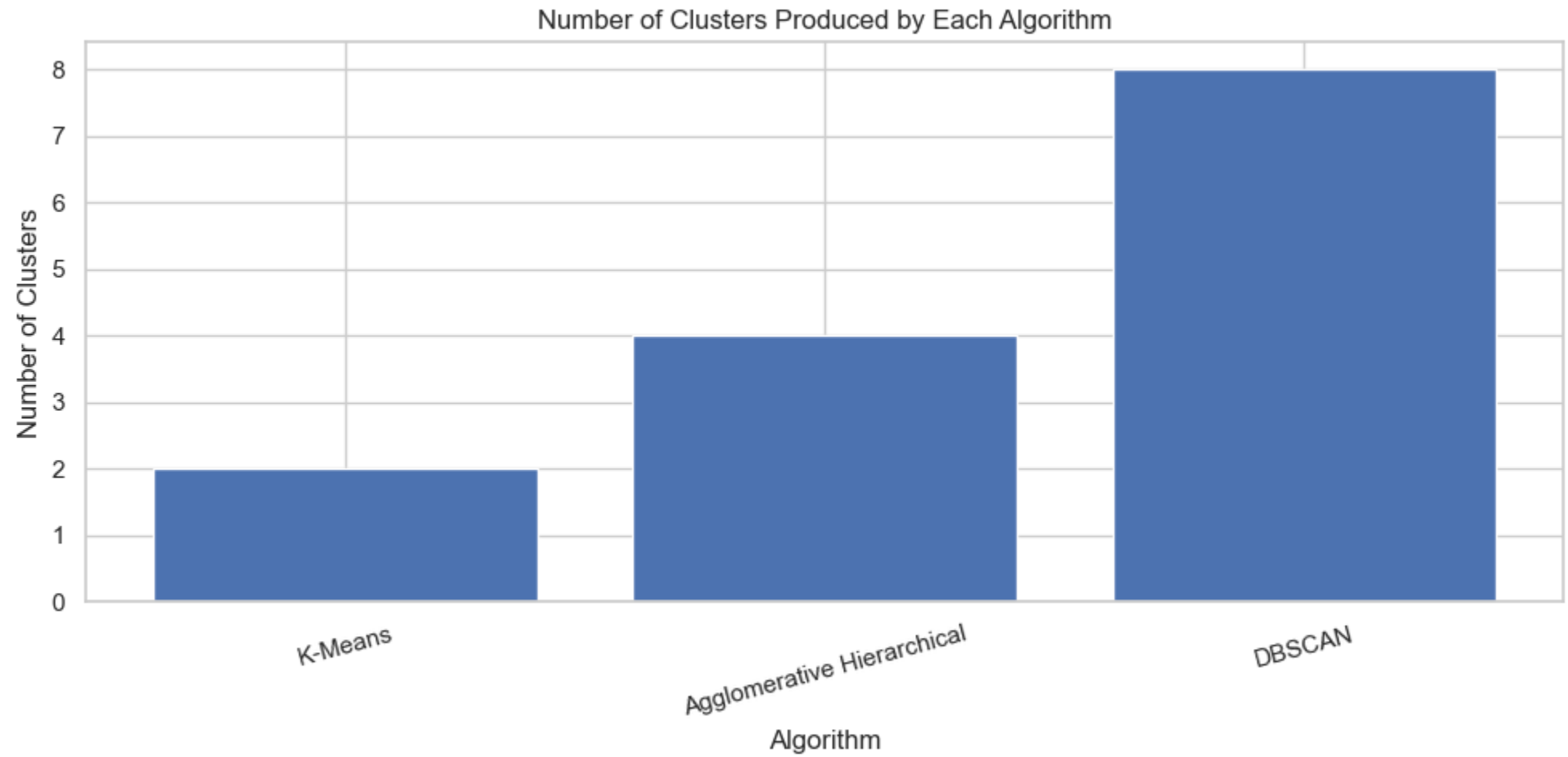


Table 1 — Generated by Notebook Cell 2

Unnamed: 0	CUST_ID	BALANCE	BALANCE	PERCENTAGE	ONEOFF_P	PURCHASE	PERCENTAGE	PURCHASE	ONEOFF_P	PURCHASE	PERCENTAGE	PURCHASE	ONEOFF_P	PURCHASE	ONEOFF_P	MINIMUM_P	PERCENTAGE	PAYMENT
0	C10001	40.9010	0.8180	95.4000	0.0000	95.4000	0.0000	0.1670	0.0000	0.0830	0.0000	0	2	1000.0000	201.8020	139.5100	0.0000	12
1	C10002	3202.4670	0.9090	0.0000	0.0000	0.0000	6442.9450	0.0000	0.0000	0.0000	0.2500	4	0	7000.0000	4103.0330	1072.3400	0.2220	12
2	C10003	2495.1490	1.0000	773.1700	773.1700	0.0000	0.0000	1.0000	1.0000	0.0000	0.0000	0	12	7500.0000	622.0670	627.2850	0.0000	12
3	C10004	1666.6710	0.6360	1499.0000	1499.0000	0.0000	205.7880	0.0830	0.0830	0.0000	0.0830	1	1	7500.0000	0.0000	nan	0.0000	12
4	C10005	817.7140	1.0000	16.0000	16.0000	0.0000	0.0000	0.0830	0.0830	0.0000	0.0000	0	1	1200.0000	678.3350	244.7910	0.0000	12

Table 2 — Generated by Notebook Cell 2

Unnamed: 0	CUST_ID	BALANCE	BALANCE	PERCENTAGE	ONEOFF_PAYMENT	PURCHASED	PERCENTAGE	PURCHASED	PERCENTAGE	PERCENTAGE	PERCENTAGE	PERCENTAGE	PERCENTAGE	PERCENTAGE	PERCENTAGE	PERCENTAGE	PERCENTAGE	PERCENTAGE
7669	C17875	16.8350	0.4550	15.0000	15.0000	0.0000	209.0250	0.0910	0.0910	0.0000	0.0910	1	1	7500.0000	430.2130	86.9600	0.0000	11
6125	C16296	540.0210	1.0000	612.2300	495.6100	116.6200	1708.9230	0.6670	0.1670	0.5000	0.3330	10	10	2000.0000	1642.0690	419.9560	0.0000	12
7028	C17219	119.2380	1.0000	342.7400	0.0000	342.7400	0.0000	1.0000	0.0000	1.0000	0.0000	0	20	2000.0000	327.1660	165.2070	0.0000	12
3019	C13108	894.0820	1.0000	1901.7100	1853.1100	48.6000	206.6190	0.6670	0.6670	0.4170	0.0830	1	33	1500.0000	947.1300	220.7450	0.0000	12
3479	C13576	1294.1450	1.0000	3059.1000	1836.9800	1222.1200	0.0000	1.0000	0.4170	1.0000	0.0000	0	42	7000.0000	5560.0340	497.6380	0.0830	12

Table 3 — Generated by Notebook Cell 2

Unnamed: 0	Data Type
CUST_ID	object
BALANCE	float64
BALANCE_FREQUENCY	float64
PURCHASES	float64
ONEOFF_PURCHASES	float64
INSTALLMENTS_PURCHASES	float64
CASH_ADVANCE	float64
PURCHASES_FREQUENCY	float64
ONEOFF_PURCHASES_FREQUENCY	float64
PURCHASES_INSTALLMENTS_FREQUENCY	float64
CASH_ADVANCE_FREQUENCY	float64
CASH_ADVANCE_TRX	int64
PURCHASES_TRX	int64
CREDIT_LIMIT	float64
PAYMENTS	float64
MINIMUM_PAYMENTS	float64
PRC_FULL_PAYMENT	float64
TENURE	int64

Table 4 — Generated by Notebook Cell 2

Unnamed: 0	Missing_Count	Missing_Percentage
0	313	3.4970
1	1	0.0110
2	0	0.0000
3	0	0.0000
4	0	0.0000
5	0	0.0000
6	0	0.0000
7	0	0.0000
8	0	0.0000
9	0	0.0000
10	0	0.0000
11	0	0.0000
12	0	0.0000
13	0	0.0000
14	0	0.0000
15	0	0.0000
16	0	0.0000
17	0	0.0000

Table 5 — Generated by Notebook Cell 2

Unnamed: 0	Infinite_Count
BALANCE	0
BALANCE_FREQUENCY	0
PURCHASES	0
ONEOFF_PURCHASES	0
INSTALLMENTS_PURCHASES	0
CASH_ADVANCE	0
PURCHASES_FREQUENCY	0
ONEOFF_PURCHASES_FREQUENCY	0
PURCHASES_INSTALLMENTS_FREQUENCY	0
CASH_ADVANCE_FREQUENCY	0
CASH_ADVANCE_TRX	0
PURCHASES_TRX	0
CREDIT_LIMIT	0
PAYMENTS	0
MINIMUM_PAYMENTS	0
PRC_FULL_PAYMENT	0
TENURE	0

Table 6 — Generated by Notebook Cell 2

Unnamed: 0	Missing_Count	Missing_Percentage
MINIMUM_PAYMENTS	313	3.4970
CREDIT_LIMIT	1	0.0110

Table 7 — Generated by Notebook Cell 2

Unnamed: 0	count	mean	std	min	25%	50%	75%	max	missing_count	missing_percentage
0	8950.0000	1564.4750	2081.5320	0.0000	128.2820	873.3850	2054.1400	19,043.14	0	0.0000
1	8950.0000	0.8770	0.2370	0.0000	0.8890	1.0000	1.0000	1.0000	0	0.0000
2	8950.0000	1003.2050	2136.6350	0.0000	39.6350	361.2800	1110.1300	49,039.57	0	0.0000
3	8950.0000	592.4370	1659.8880	0.0000	0.0000	38.0000	577.4050	40,761.25	0	0.0000
4	8950.0000	411.0680	904.3380	0.0000	0.0000	89.0000	468.6370	22,500.00	0	0.0000
5	8950.0000	978.8710	2097.1640	0.0000	0.0000	0.0000	1113.8210	47,137.21	0	0.0000
6	8950.0000	0.4900	0.4010	0.0000	0.0830	0.5000	0.9170	1.0000	0	0.0000
7	8950.0000	0.2020	0.2980	0.0000	0.0000	0.0830	0.3000	1.0000	0	0.0000
8	8950.0000	0.3640	0.3970	0.0000	0.0000	0.1670	0.7500	1.0000	0	0.0000
9	8950.0000	0.1350	0.2000	0.0000	0.0000	0.0000	0.2220	1.5000	0	0.0000
10	8950.0000	3.2490	6.8250	0.0000	0.0000	0.0000	4.0000	123.0000	0	0.0000
11	8950.0000	14.7100	24.8580	0.0000	1.0000	7.0000	17.0000	358.0000	0	0.0000
12	8949.0000	4494.4490	3638.8160	50.0000	1600.0000	3000.0000	6500.0000	30,000.00	1	0.0110
13	8950.0000	1733.1440	2895.0640	0.0000	383.2760	856.9020	1901.1340	50,721.48	0	0.0000
14	8637.0000	864.2070	2372.4470	0.0190	169.1240	312.3440	825.4850	76,406.21	313	3.4970
15	8950.0000	0.1540	0.2920	0.0000	0.0000	0.0000	0.1430	1.0000	0	0.0000
16	8950.0000	11.5170	1.3380	6.0000	12.0000	12.0000	12.0000	12.0000	0	0.0000

Table 8 — Generated by Notebook Cell 2

Unnamed: 0	Feature	Minimum	Maximum	Invalid_Outside_0_1	Status
0	BALANCE_FREQUENCY	0.0000	1.0000	0	Valid
1	PURCHASES_FREQUENCY	0.0000	1.0000	0	Valid
2	ONEOFF_PURCHASES_FREQUENCY	0.0000	1.0000	0	Valid
3	PURCHASES_INSTALLMENTS_FREQUENCY	0.0000	1.0000	0	Valid
4	CASH_ADVANCE_FREQUENCY	0.0000	1.5000	8	Review Required
5	PRC_FULL_PAYMENT	0.0000	1.0000	0	Valid

Table 9 — Generated by Notebook Cell 2

Unnamed: 0	PURCHASES	ONEOFF_PURCHASES	INSTALLMENTS_PURCHASES	Calculated_Total	Absolute_Difference
0	5629.4100	0.0000	6229.4100	6229.4100	600.0000
1	205.0600	0.0000	607.7600	607.7600	402.7000
2	279.7600	0.0000	578.5500	578.5500	298.7900
3	339.1100	611.6500	12.4100	624.0600	284.9500
4	510.0000	0.0000	780.0000	780.0000	270.0000
5	426.2500	0.0000	653.5500	653.5500	227.3000
6	486.2700	580.2000	0.0000	580.2000	93.9300
7	400.4100	0.0000	489.3900	489.3900	88.9800
8	0.0000	0.0000	66.9500	66.9500	66.9500
9	3393.2500	3364.5900	77.6600	3442.2500	49.0000

Table 10 — Generated by Notebook Cell 2

Unnamed: 0	Feature	Minimum	Maximum	Negative_Count	Zero_Count	Negative_Percentage	Zero_Percentage
0	BALANCE	0.0000	19,043.14	0	80	0.0000	0.8940
1	PURCHASES	0.0000	49,039.57	0	2044	0.0000	22.8380
2	ONEOFF_PURCHASES	0.0000	40,761.25	0	4302	0.0000	48.0670
3	INSTALLMENTS_PURCHASES	0.0000	22,500.00	0	3916	0.0000	43.7540
4	CASH_ADVANCE	0.0000	47,137.21	0	4628	0.0000	51.7090
5	CREDIT_LIMIT	50.0000	30,000.00	0	0	0.0000	0.0000
6	PAYMENTS	0.0000	50,721.48	0	240	0.0000	2.6820
7	MINIMUM_PAYMENTS	0.0190	76,406.21	0	0	0.0000	0.0000
8	PRC_FULL_PAYMENT	0.0000	1.0000	0	5903	0.0000	65.9550

Table 11 — Generated by Notebook Cell 2

Unnamed: 0	Feature	Minimum	Maximum	Negative_Count	Zero_Count
0	PURCHASES_TRX	0.0000	358.0000	0	2044
1	CASH_ADVANCE_TRX	0.0000	123.0000	0	4628
2	INSTALLMENTS_PURCHASES	0.0000	22,500.00	0	3916

Table 12 — Generated by Notebook Cell 2

Unnamed: 0	Customer_Count	Percentage
0	204	2.2790
1	190	2.1230
2	196	2.1900
3	175	1.9550
4	236	2.6370
5	365	4.0780
6	7584	84.7370

Table 13 — Generated by Notebook Cell 3

Unnamed: 0	count	mean	median	std	min	max	missing_count	missing_percent	skewness
BALANCE	8950.0000	1564.4750	873.3850	2081.5320	0.0000	19,043.14	0	0.0000	2.3930
BALANCE_FREQUENCY	8950.0000	0.8770	1.0000	0.2370	0.0000	1.0000	0	0.0000	-2.0230
PURCHASES	8950.0000	1003.2050	361.2800	2136.6350	0.0000	49,039.57	0	0.0000	8.1440
ONEOFF_PURCHASES	8950.0000	592.4370	38.0000	1659.8880	0.0000	40,761.25	0	0.0000	10.0450
INSTALLMENTS_PURCHASED	8950.0000	411.0680	89.0000	904.3380	0.0000	22,500.00	0	0.0000	7.2990
CASH_ADVANCE	8950.0000	978.8710	0.0000	2097.1640	0.0000	47,137.21	0	0.0000	5.1670
PURCHASES_FREQUENCY	8950.0000	0.4900	0.5000	0.4010	0.0000	1.0000	0	0.0000	0.0600
ONEOFF_PURCHASES_FREQUENCY	8950.0000	0.2020	0.0830	0.2980	0.0000	1.0000	0	0.0000	1.5360
PURCHASES_INSTALLMENTS_FREQUENCY	8950.0000	0.3640	0.1670	0.3970	0.0000	1.0000	0	0.0000	0.5090
CASH_ADVANCE_FREQUENCY	8950.0000	0.1350	0.0000	0.2000	0.0000	1.5000	0	0.0000	1.8290
CASH_ADVANCE_TRX	8950.0000	3.2490	0.0000	6.8250	0.0000	123.0000	0	0.0000	5.7210
PURCHASES_TRX	8950.0000	14.7100	7.0000	24.8580	0.0000	358.0000	0	0.0000	4.6310
CREDIT_LIMIT	8949.0000	4494.4490	3000.0000	3638.8160	50.0000	30,000.00	1	0.0110	1.5220
PAYMENTS	8950.0000	1733.1440	856.9020	2895.0640	0.0000	50,721.48	0	0.0000	5.9080
MINIMUM_PAYMENTS	8637.0000	864.2070	312.3440	2372.4470	0.0190	76,406.21	313	3.4970	13.6230
PRC_FULL_PAYMENT	8950.0000	0.1540	0.0000	0.2920	0.0000	1.0000	0	0.0000	1.9430
TENURE	8950.0000	11.5170	12.0000	1.3380	6.0000	12.0000	0	0.0000	-2.9430

Table 14 — Generated by Notebook Cell 3

Unnamed: 0	skewness
MINIMUM_PAYMENTS	13.6230
ONEOFF_PURCHASES	10.0450
PURCHASES	8.1440
INSTALLMENTS_PURCHASES	7.2990
PAYMENTS	5.9080
CASH_ADVANCE_TRX	5.7210
CASH_ADVANCE	5.1670
PURCHASES_TRX	4.6310
TENURE	-2.9430
BALANCE	2.3930
BALANCE_FREQUENCY	-2.0230

Table 15 — Generated by Notebook Cell 3

('Unnamed: 0_level_0', 'feature')	('missing_count', 'Unnamed: 1_level_1')	('missing_percent', 'Unnamed: 2_level_1')
MINIMUM_PAYMENTS	313	3.4970
CREDIT_LIMIT	1	0.0110

Table 16 — Generated by Notebook Cell 3

('Unnamed: 0_level_0', 'feature')	('Q1', 'Unnamed: 1_level_1')	('Q3', 'Unnamed: 2_level_1')	('IQR', 'Unnamed: 3_level_1')	('lower_bound', 'Unnamed: 4_level_1')	('upper_bound', 'Unnamed: 5_level_1')	('flagged_count', 'Unnamed: 6_level_1')	('flagged_percent', 'Unnamed: 7_level_1')
BALANCE	128.2820	2054.1400	1925.8580	-2760.5050	4942.9270	695	7.7650
PURCHASES	39.6350	1110.1300	1070.4950	-1566.1080	2715.8730	808	9.0280
ONEOFF_PURCHASES	0.0000	577.4050	577.4050	-866.1080	1443.5120	1013	11.3180
INSTALLMENTS_PURCHASES	50.0000	468.6380	468.6380	-702.9560	1171.5940	867	9.6870
CASH_ADVANCE	0.0000	1113.8210	1113.8210	-1670.7320	2784.5530	1030	11.5080
PAYMENTS	383.2760	1901.1340	1517.8580	-1893.5110	4177.9220	808	9.0280
MINIMUM_PAYMENTS	169.1240	825.4850	656.3620	-815.4190	1810.0280	841	9.7370
CREDIT_LIMIT	1600.0000	6500.0000	4900.0000	-5750.0000	13,850.00	248	2.7710

Table 17 — Generated by Notebook Cell 3

('Unnamed: 0_level_0', 'feature')	('negative_count', 'Unnamed: 1_level_1')	('negative_percent', 'Unnamed: 2_level_1')
BALANCE	0	0.0000
PURCHASES	0	0.0000
ONEOFF_PURCHASES	0	0.0000
INSTALLMENTS_PURCHASES	0	0.0000
CASH_ADVANCE	0	0.0000
PAYMENTS	0	0.0000
MINIMUM_PAYMENTS	0	0.0000
CREDIT_LIMIT	0	0.0000

Table 18 — Generated by Notebook Cell 3

Unnamed: 0	valid_count	zero_count	zero_percent	nonzero_count
PRC_FULL_PAYMENT	8950	5903	65.9550	3047
CASH_ADVANCE_FREQUENCY	8950	4628	51.7090	4322
CASH_ADVANCE	8950	4628	51.7090	4322
CASH_ADVANCE_TRX	8950	4628	51.7090	4322
ONEOFF_PURCHASES_FREQUENCY	8950	4302	48.0670	4648
ONEOFF_PURCHASES	8950	4302	48.0670	4648
INSTALLMENTS_PURCHASES	8950	3916	43.7540	5034
PURCHASES_INSTALLMENTS_FREQUENCY	8950	3915	43.7430	5035
PURCHASES	8950	2044	22.8380	6906
PURCHASES_TRX	8950	2044	22.8380	6906
PURCHASES_FREQUENCY	8950	2043	22.8270	6907
PAYMENTS	8950	240	2.6820	8710
BALANCE	8950	80	0.8940	8870
BALANCE_FREQUENCY	8950	80	0.8940	8870
CREDIT_LIMIT	8949	0	0.0000	8949
MINIMUM_PAYMENTS	8637	0	0.0000	8637
TENURE	8950	0	0.0000	8950

Table 19 — Generated by Notebook Cell 3

('Unnamed: 0_level_0', 'TENURE')	('customer_count', 'Unnamed: 1_level_1')	('percentage', 'Unnamed: 2_level_1')
6	204	2.2790
7	190	2.1230
8	196	2.1900
9	175	1.9550
10	236	2.6370
11	365	4.0780
12	7584	84.7370

Table 20 — Generated by Notebook Cell 3

('Unnamed: 0_level_0', 'feature')	('minimum', 'Unnamed: 1_level_1')	('maximum', 'Unnamed: 2_level_1')	('below_zero_count', 'Unnamed: 3_level_1')	('above_one_count', 'Unnamed: 4_level_1')	('outside_0_1_count', 'Unnamed: 5_level_1')
BALANCE_FREQUENCY	0.0000	1.0000	0	0	0
PURCHASES_FREQUENCY	0.0000	1.0000	0	0	0
ONEOFF_PURCHASES_FREQUENCY	0.0000	1.0000	0	0	0
PURCHASES_INSTALLMENTS_FREQUENCY	0.0000	1.0000	0	0	0
CASH_ADVANCE_FREQUENCY	0.0000	1.5000	0	8	8
PRC_FULL_PAYMENT	0.0000	1.0000	0	0	0

Table 21 — Generated by Notebook Cell 3

Unnamed: 0	source_row_index	CASH_ADVANCE_FREQUENCY
681	681	1.2500
1626	1626	1.1670
2555	2555	1.1250
2608	2608	1.1000
3038	3038	1.5000
3253	3253	1.1670
8055	8055	1.0910
8365	8365	1.1430

Table 22 — Generated by Notebook Cell 3

Unnamed: 0	purchase_credit_utilization
count	8949.0000
mean	0.2630
std	0.4370
min	0.0000
25%	0.0110
50%	0.1170
75%	0.3380
max	8.5910
median	0.1170

Table 23 — Generated by Notebook Cell 3

Unnamed: 0	CASH_ADVANCE	PURCHASES	CREDIT_LIMIT	BALANCE	CASH_ADVANCE_TRX	PURCHASES_TRX
2159	47,137.21	431.9300	19,600.00	10,905.05	123	21
1059	29,282.11	3719.0000	15,500.00	8823.2840	26	61
71	27,296.49	4523.2700	7000.0000	2990.4220	27	33
7254	26,268.70	1750.6600	8500.0000	4530.2050	10	36
7645	26,194.05	0.0000	9000.0000	7081.1710	69	0
2454	23,130.82	0.0000	15,000.00	10,915.55	23	0
883	22,665.78	0.0000	18,500.00	14,581.46	30	0
6371	21,943.85	0.0000	13,000.00	4529.8960	12	0
3806	20,712.67	4995.8000	16,500.00	14,100.25	36	34
6455	20,277.33	1490.7500	11,000.00	9110.9690	22	5

Table 24 — Generated by Notebook Cell 3

Unnamed: 0	source_row_index	PURCHASES	ONEOFF_PURCHASES	INSTALLMENTS_PURCHASES	component_sum	signed_difference	absolute_difference
6857	6857	5629.4100	0.0000	6229.4100	6229.4100	-600.0000	600.0000
5231	5231	205.0600	0.0000	607.7600	607.7600	-402.7000	402.7000
5967	5967	279.7600	0.0000	578.5500	578.5500	-298.7900	298.7900
6803	6803	339.1100	611.6500	12.4100	624.0600	-284.9500	284.9500
8834	8834	510.0000	0.0000	780.0000	780.0000	-270.0000	270.0000
6790	6790	426.2500	0.0000	653.5500	653.5500	-227.3000	227.3000
5846	5846	486.2700	580.2000	0.0000	580.2000	-93.9300	93.9300
510	510	400.4100	0.0000	489.3900	489.3900	-88.9800	88.9800
5737	5737	0.0000	0.0000	66.9500	66.9500	-66.9500	66.9500
908	908	3393.2500	3364.5900	77.6600	3442.2500	-49.0000	49.0000

Table 25 — Generated by Notebook Cell 3

Unnamed: 0	BALANCE	BALANCE	FREQUENCY	ONEOFF_PURCHASES	ONEOFF_PURCHASES	ONEOFF_PURCHASES	ONEOFF_PURCHASES	ONEOFF_PURCHASES	ONEOFF_PURCHASES	ONEOFF_PURCHASES	ONEOFF_PURCHASES	ONEOFF_PURCHASES	ONEOFF_PURCHASES	CREDIT_LIMIT	PAYMENTS	MINIMUM_PAYMENT	PRC_FULL_PAYMENT	TENURE
BALANCE	1.0000	0.3200	0.1800	0.1600	0.1300	0.5000	-0.0800	0.0700	-0.0600	0.4500	0.3900	0.1500	0.5300	0.3200	0.4000	-0.3200	0.0700	
BALANCE_FREQUENCY	1.0000	0.1300	0.1000	0.1200	0.1000	0.2300	0.2000	0.1800	0.1900	0.1400	0.1900	0.1000	0.0700	0.1300	-0.1000	0.1200		
PURCHASES	0.1800	0.1300	1.0000	0.9200	0.6800	-0.0500	0.3900	0.5000	0.3200	-0.1200	-0.0700	0.6900	0.3600	0.6000	0.0900	0.1800	0.0900	
ONEOFF_PURCHASES	0.1000	0.9200	1.0000	0.3300	-0.0300	0.2600	0.5200	0.1300	-0.0800	-0.0500	0.5500	0.3200	0.5700	0.0500	0.1300	0.0600		
INSTALLMENTS	0.3000	0.6800	0.3300	1.0000	-0.0600	0.4400	0.2100	0.5100	-0.1300	-0.0700	0.6300	0.2600	0.3800	0.1300	0.1800	0.0900		
CASH_ADVANCE	0.1000	0.1000	-0.0500	-0.0300	-0.0600	1.0000	-0.2200	-0.0900	-0.1800	0.6300	0.6600	-0.0800	0.3000	0.4500	0.1400	-0.1500	-0.0700	
PURCHASES_FREQUENCY	0.2300	0.3900	0.2600	0.4400	-0.2200	1.0000	0.5000	0.8600	-0.3100	-0.2000	0.5700	0.1200	0.1000	0.0000	0.3100	0.0600		
ONEOFF_PURCHASES_FREQUENCY	0.5000	0.5200	0.2100	-0.0900	0.5000	1.0000	0.1400	-0.1100	-0.0700	0.5400	0.3000	0.2400	-0.0300	0.1600	0.0800			
PURCHASES_INSTALLMENTS	0.1300	0.5100	-0.1800	0.8600	0.1400	1.0000	-0.2600	-0.1700	0.5300	0.0600	0.0900	0.0300	0.2500	0.0700				
CASH_ADVANCE_FREQUENCY	0.1300	-0.1200	-0.0800	-0.1300	0.6300	-0.3100	-0.1100	-0.2600	1.0000	0.8000	-0.1300	0.1300	0.1800	0.1000	-0.2500	-0.1300		
CASH_ADVANCE_INSTALLMENTS	0.1400	-0.0700	-0.0500	-0.0700	0.6600	-0.2000	-0.0700	-0.1700	0.8000	1.0000	-0.0700	0.1500	0.2600	0.1100	-0.1700	-0.0400		
PURCHASES_TENURE	0.1900	0.6900	0.5500	0.6300	-0.0800	0.5700	0.5400	0.5300	-0.1300	-0.0700	1.0000	0.2700	0.3700	0.1000	0.1600	0.1200		
CREDIT_LIMIT	0.5300	0.1000	0.3600	0.3200	0.2600	0.3000	0.1200	0.3000	0.0600	0.1300	0.1500	0.2700	1.0000	0.4200	0.1300	0.0600	0.1400	
PAYMENTS	0.3200	0.0700	0.6000	0.5700	0.3800	0.4500	0.1000	0.2400	0.0900	0.1800	0.2600	0.3700	0.4200	1.0000	0.1300	0.1100	0.1100	
MINIMUM_PAYMENTS	0.1300	0.0900	0.0500	0.1300	0.1400	0.0000	-0.0300	0.0300	0.1000	0.1100	0.1000	0.1300	0.1300	1.0000	-0.1400	0.0600		
PRC_FULL_PAYMENT	-0.1000	0.1800	0.1300	0.1800	-0.1500	0.3100	0.1600	0.2500	-0.2500	-0.1700	0.1600	0.0600	0.1100	-0.1400	1.0000	-0.0200		
TENURE	0.0700	0.1200	0.0900	0.0600	0.0900	-0.0700	0.0600	0.0800	0.0700	-0.1300	-0.0400	0.1200	0.1400	0.1100	0.0600	-0.0200	1.0000	

Table 26 — Generated by Notebook Cell 3

Unnamed: 0	feature_1	feature_2	correlation	absolute_correlation
0	PURCHASES	ONEOFF_PURCHASES	0.9170	0.9170
1	PURCHASES_FREQUENCY	PURCHASES_INSTALLMENTS_FREQUENCY	0.8630	0.8630

Table 27 — Generated by Notebook Cell 3

Unnamed: 0	feature_1	feature_2	correlation	absolute_correlation
0	CASH_ADVANCE_FREQUENCY	CASH_ADVANCE_TRX	0.9830	0.9830
1	ONEOFF_PURCHASES	ONEOFF_PURCHASES_FREQUENCY	0.9520	0.9520
2	CASH_ADVANCE	CASH_ADVANCE_TRX	0.9520	0.9520
3	CASH_ADVANCE	CASH_ADVANCE_FREQUENCY	0.9410	0.9410
4	PURCHASES_FREQUENCY	PURCHASES_TRX	0.9240	0.9240
5	INSTALLMENTS_PURCHASES	PURCHASES_INSTALLMENTS_FREQUENCY	0.9230	0.9230
6	BALANCE	MINIMUM_PAYMENTS	0.9000	0.9000
7	PURCHASES	PURCHASES_TRX	0.8850	0.8850
8	PURCHASES_FREQUENCY	PURCHASES_INSTALLMENTS_FREQUENCY	0.8520	0.8520

Table 28 — Generated by Notebook Cell 3

Unnamed: 0	finding	evidence	recommended_next_action
0	Missing numeric values	314 missing cells; largest affected feature: M...	Choose a justified missing-value strategy duri...
1	Infinite values	0 infinite numeric cells	Investigate the source of any infinities befor...
2	Skewed features	11 features meet the skewness threshold	Review distributions and consider transformati...
3	Negative financial amounts	0 negative amount cells across listed features	Check feature definitions and data validity; d...
4	Potential extreme values	6310 feature-level flags (the same customer ma...	Review the distributions and retain valid cust...
5	Features with many zeros	4 features have at least 50% zeros among non-m...	Determine whether zeros indicate meaningful in...
6	Frequency/proportion range	8 values outside 0–1 in checked columns	Verify variable definitions and investigate un...
7	Purchase-component consistency	19 of 8950 complete rows exceed tolerance	Verify definitions and flagged records before ...
8	Pearson correlation	2 pairs meet the absolute threshold	Review overlapping features in light of the cl...
9	Spearman correlation	9 pairs meet the absolute threshold	Compare rank-based relationships with Pearson ...
10	Zero-purchase customers	2,044 of 8,950 valid PURCHASES observations ar...	Consider zero-purchase behaviour as a potentia...
11	Purchase-to-credit utilization	Rough PURCHASES/CREDIT_LIMIT ratio was availa...	Review this ratio as a behavioural feature dur...

Table 29 — Generated by Notebook Cell 5

Unnamed: 0	0
CREDIT_LIMIT	1
MINIMUM_PAYMENTS	313

Table 30 — Generated by Notebook Cell 5

Unnamed: 0	feature	missing_before	method	imputation_value
0	CREDIT_LIMIT	1	Median	3000.0000
1	MINIMUM_PAYMENTS	313	Median	312.3440

Table 31 — Generated by Notebook Cell 5

Unnamed: 0	feature	minimum	maximum	outside_0_1_count
0	BALANCE_FREQUENCY	0.0000	1.0000	0
1	PURCHASES_FREQUENCY	0.0000	1.0000	0
2	ONEOFF_PURCHASES_FREQUENCY	0.0000	1.0000	0
3	PURCHASES_INSTALLMENTS_FREQUENCY	0.0000	1.0000	0
4	CASH_ADVANCE_FREQUENCY	0.0000	1.5000	8
5	PRC_FULL_PAYMENT	0.0000	1.0000	0

Table 32 — Generated by Notebook Cell 5

Unnamed: 0	feature	Q1	Q3	IQR	lower_bound	upper_bound	lower_capped	upper_capped	total_capped
0	BALANCE	128.2820	2054.1400	1925.8580	-5649.2920	7831.7140	0	194	194
1	PURCHASES	39.6350	1110.1300	1070.4950	-3171.8500	4321.6150	0	391	391
2	ONEOFF_PURCHASES	0.0000	577.4050	577.4050	-1732.2150	2309.6200	0	566	566
3	INSTALLMENTS_PURCHASES	0.0000	468.6380	468.6380	-1405.9120	1874.5500	0	395	395
4	CASH_ADVANCE	0.0000	1113.8210	1113.8210	-3341.4630	4455.2850	0	486	486
5	PAYMENTS	383.2760	1901.1340	1517.8580	-4170.2980	6454.7090	0	402	402
6	MINIMUM_PAYMENTS	170.8580	788.7140	617.8560	-1682.7100	2642.2810	0	470	470
7	CREDIT_LIMIT	1600.0000	6500.0000	4900.0000	-13,100.00	21,200.00	0	10	10

Table 33 — Generated by Notebook Cell 5

Unnamed: 0	skewness
CASH_ADVANCE_TRX	5.7210
PURCHASES_TRX	4.6310
TENURE	-2.9430
BALANCE_FREQUENCY	-2.0230
PRC_FULL_PAYMENT	1.9430
CASH_ADVANCE_FREQUENCY	1.8290
PURCHASES	1.7910
INSTALLMENTS_PURCHASES	1.7570
PAYMENTS	1.7350
BALANCE	1.7330
MINIMUM_PAYMENTS	1.7230
ONEOFF_PURCHASES	1.7090
CASH_ADVANCE	1.6880
ONEOFF_PURCHASES_FREQUENCY	1.5360
CREDIT_LIMIT	1.4540
PURCHASES_INSTALLMENTS_FREQUENCY	0.5090
PURCHASES_FREQUENCY	0.0600

Table 34 — Generated by Notebook Cell 5

Unnamed: 0	minimum	median	maximum	missing
PURCHASE_TO_CREDIT_RATIO	0.0000	0.1170	8.1240	0
CASH_ADVANCE_TO_CREDIT_RATIO	0.0000	0.0000	89.1060	0
ONEOFF_PURCHASE_SHARE	0.0000	0.1120	1.8040	0
INSTALLMENT_PURCHASE_SHARE	0.0000	0.1930	2.9640	0

Table 35 — Generated by Notebook Cell 5

Unnamed: 0	feature	maximum	99th_percentile
0	PURCHASE_TO_CREDIT_RATIO	8.1240	1.6680
1	CASH_ADVANCE_TO_CREDIT_RATIO	89.1060	1.4850
2	ONEOFF_PURCHASE_SHARE	1.8040	1.0000
3	INSTALLMENT_PURCHASE_SHARE	2.9640	1.0000

Table 36 — Generated by Notebook Cell 5

Unnamed: 0	mean	std
BALANCE	-0.0000	1.0000
BALANCE_FREQUENCY	0.0000	1.0000
PURCHASES	0.0000	1.0000
PURCHASES_FREQUENCY	-0.0000	1.0000
PURCHASES_TRX	0.0000	1.0000
CASH_ADVANCE	-0.0000	1.0000
CASH_ADVANCE_FREQUENCY	0.0000	1.0000
CASH_ADVANCE_TRX	-0.0000	1.0000
CREDIT_LIMIT	0.0000	1.0000
PAYMENTS	0.0000	1.0000
MINIMUM_PAYMENTS	0.0000	1.0000
PRC_FULL_PAYMENT	0.0000	1.0000
PURCHASE_TO_CREDIT_RATIO	0.0000	1.0000
ONEOFF_PURCHASE_SHARE	-0.0000	1.0000
INSTALLMENT_PURCHASE_SHARE	-0.0000	1.0000

Table 37 — Generated by Notebook Cell 5

Unnamed: 0	k	inertia	silhouette_score
0	2	105,285.96	0.2260
1	3	90,529.07	0.1960
2	4	78,994.79	0.2060
3	5	71,132.91	0.2190
4	6	66,637.19	0.2110
5	7	63,069.31	0.2120
6	8	59,554.19	0.2060
7	9	56,978.39	0.2120
8	10	54,348.86	0.2160

Table 38 — Generated by Notebook Cell 5

Unnamed: 0	KMeans_Cluster	customer_count	percentage
0	0	5580	62.3460
1	1	3370	37.6540

Table 39 — Generated by Notebook Cell 5

(Unnamed: 0_level_0', 'KMERCHANDISE', 'Unnamed: 1_level_0', 'PURCHASES', 'Unnamed: 2_level_0', 'CASH ADVANCE', 'Unnamed: 3_level_0', 'REDEMPTION', 'Unnamed: 4_level_0', 'PAYMENTS', 'Unnamed: 5_level_0', 'TENDR', 'Unnamed: 6_level_0', 'CUSTOMER_COUNT', 'Unnamed: 7_level_0', 'CUSTOMER_PERCENT', 'Unnamed: 8_level_0')								
0	888.6940	1391.7120	120.2410	4309.1770	1510.2990	11.6070	5580	62.3460
1	2683.4230	359.9200	2400.5790	4801.3120	2102.1280	11.3680	3370	37.6540

Table 40 — Generated by Notebook Cell 5

0	1	2	3	4	5	6	7	8
0	0.8540	0.6770	0.2690	0.5170	0.0270	0.2250	20.3160	0.4690
1	0.9150	0.1810	0.0920	0.1130	0.3140	0.0360	5.4260	7.8520

Table 41 — Generated by Notebook Cell 5

(Unnamed: 0_level_0', 'KMeans_Cluster')	(PURCHASE_TO_CREDIT_RATIO', 'Unnamed: 1_level_0')	(ONEOFF_PURCHASE_SHARE', 'Unnamed: 2_level_0')	(INSTALLMENT_PURCHASE_SHARE', 'Unnamed: 3_level_1')
0	0.3550	0.4290	0.5430
1	0.0610	0.2650	0.1280

Table 42 — Generated by Notebook Cell 5

Unnamed: 0	KMeans_Cluster	BALANCE	PURCHASES	CASH_ADVANCE	CREDIT_LIMIT	PAYMENTS	TENURE	CUSTOMER_COUNT	CUSTOMER_PERCENT	Business_Persona	Key_Characteristics
0	0	888.6940	1391.7120	120.2410	4309.1770	1510.2990	11.6070	5580	62.3460	Transactors	high purchase activity, higher full-paym
1	1	2683.4230	359.9200	2400.5790	4801.3120	2102.1280	11.3680	3370	37.6540	Cash-Advance Reliant	high cash-advance activity, above-aver

Table 43 — Generated by Notebook Cell 5

Unnamed: 0	KMeans_Cluster	BALANCE	PURCHASES	PURCHASES_FREQUENCY	CASH_ADVANCE	CASH_ADVANCE_FREQUENCY	CASH_LIMIT	PRC_FULL_PAYMENT	Business_Persona	Key_Characteristics
0	0	888.6940	1391.7120	0.6770	120.2410	0.0270	4309.1770	0.2250	Transactors	high purchase activity, higher full-payme
1	1	2683.4230	359.9200	0.1810	2400.5790	0.3140	4801.3120	0.0360	Cash-Advance Reliant	high cash-advance activity, above-avera

Table 44 — Generated by Notebook Cell 6

Unnamed: 0	TARGET_CLUSTERS	LOWER_MERGE_DISTANCE	UPPER_MERGE_DISTANCE	CUT_DISTANCE	RESULTING_CLUSTERS
0	3	137.4900	168.0520	152.7710	3
1	4	130.2360	137.4900	133.8630	4
2	5	90.4680	130.2360	110.3520	5

Table 45 — Generated by Notebook Cell 6

Unnamed: 0	LINKAGE	N_CLUSTERS	SILHOUETTE_SCORE
0	average	4	0.3180
1	complete	4	0.1940
2	ward	4	0.1680

Table 46 — Generated by Notebook Cell 6

Unnamed: 0	Agg_Cluster	CUSTOMER_COUNT	CUSTOMER_PERCENT
0	0	8936	99.8440
1	1	12	0.1340
2	2	1	0.0110
3	3	1	0.0110

Table 47 — Generated by Notebook Cell 6

(Unnamed: 0_level: 1)	BALANCE_CUST	BALANCE_CUST	PURCHASES	PAID_BY	REVENUES	REVENUES	REVENUES	REVENUES	REVENUES	REVENUES	REVENUES	REVENUES	REVENUES
0	1566.5730	0.8770	1000.5700	0.4900	14.7000	980.3660	0.1350	3.2530	4500.2660	1731.8580	863.8430	0.1530	11.5170
1	256.4860	0.9680	2777.2560	0.8240	18.4170	29.1870	0.0390	0.7500	495.8330	2656.2510	1256.8280	0.3970	11.6670
2	42.5780	0.9090	4061.9600	0.9170	35.0000	0.0000	0.0000	0.0000	500.0000	3716.8810	92.0590	0.7500	12.0000
3	30.6930	0.4550	205.0600	0.9170	35.0000	0.0000	0.0000	0.0000	4500.0000	163.9820	55.6610	0.0000	12.0000

Table 48 — Generated by Notebook Cell 6

Unnamed: 0	Agg_Cluster	Business_Persona	Key_Characteristics
0	0	Mixed Behaviour Customers	mixed customer behaviour
1	1	Purchase-Oriented Customers	high purchase activity, frequent purchasing, h...
2	2	Purchase-Oriented Customers	high purchase activity, frequent purchasing, h...
3	3	Purchase-Oriented Customers	low purchase activity, frequent purchasing

Table 49 — Generated by Notebook Cell 6

Unnamed: 0	KMEANS_CUSTOMER_COUNT	AGG_CUSTOMER_COUNT
0	5580.0000	8936
1	3370.0000	12
2	nan	1
3	nan	1

Table 50 — Generated by Notebook Cell 6

('Agg_Cluster', 'KMeans_Cluster')	('0', 'Unnamed: 1_level_1')	('1', 'Unnamed: 2_level_1')	('2', 'Unnamed: 3_level_1')	('3', 'Unnamed: 4_level_1')
0	5566	12	1	1
1	3370	0	0	0

Table 51 — Generated by Notebook Cell 7

Unnamed: 0	STATISTIC	5_NN_DISTANCE
0	Minimum	0.0510
1	25th Percentile	0.6150
2	Median	0.9300
3	75th Percentile	1.3300
4	Maximum	8.2590

Table 52 — Generated by Notebook Cell 7

Unnamed: 0	EPS	MIN_SAMPLES	CLUSTERS	NOISE_COUNT	NOISE_PERCENT	NON_NOISE_CUSTOMERS	SILHOUETTE
0	0.3000	3	98	8055	90.0000	895	0.2330
1	0.5000	3	114	6489	72.5030	2461	-0.1060
2	0.7000	3	148	4797	53.5980	4153	-0.2810
3	1.0000	3	68	2676	29.8990	6274	-0.2850
4	1.5000	3	28	877	9.7990	8073	-0.0230
5	0.3000	5	31	8394	93.7880	556	0.2700
6	0.5000	5	28	7016	78.3910	1934	0.1430
7	0.7000	5	34	5472	61.1400	3478	-0.0380
8	1.0000	5	26	3170	35.4190	5780	-0.1500
9	1.5000	5	8	1101	12.3020	7849	0.1180
10	0.3000	8	11	8592	96.0000	358	0.5200
11	0.5000	8	17	7417	82.8710	1533	0.2760
12	0.7000	8	11	5964	66.6370	2986	0.1640
13	1.0000	8	10	3695	41.2850	5255	-0.0710
14	1.5000	8	5	1332	14.8830	7618	0.1630
15	0.3000	10	8	8641	96.5470	309	0.5640
16	0.5000	10	12	7635	85.3070	1315	0.2980
17	0.7000	10	10	6182	69.0730	2768	0.1760
18	1.0000	10	7	3933	43.9440	5017	0.1850
19	1.5000	10	5	1452	16.2240	7498	0.1250

Table 53 — Generated by Notebook Cell 7

Unnamed: 0	EPS	MIN_SAMPLES	CLUSTERS	NOISE_COUNT	NOISE_PERCENT	NON_NOISE_CUSTOMERS	SILHOUETTE
15	0.3000	10	8	8641	96.5470	309	0.5640
10	0.3000	8	11	8592	96.0000	358	0.5200
16	0.5000	10	12	7635	85.3070	1315	0.2980
11	0.5000	8	17	7417	82.8710	1533	0.2760
5	0.3000	5	31	8394	93.7880	556	0.2700
0	0.3000	3	98	8055	90.0000	895	0.2330
18	1.0000	10	7	3933	43.9440	5017	0.1850
17	0.7000	10	10	6182	69.0730	2768	0.1760
12	0.7000	8	11	5964	66.6370	2986	0.1640
14	1.5000	8	5	1332	14.8830	7618	0.1630

Table 54 — Generated by Notebook Cell 7

Unnamed: 0	METRIC	VALUE
0	eps	0.3000
1	min_samples	10.0000
2	Total Customers	8950.0000
3	DBSCAN Clusters	8.0000
4	Noise Customers	8641.0000
5	Noise Percentage	96.5470
6	Non-Noise Customers	309.0000
7	Non-Noise Silhouette	0.5640

Table 55 — Generated by Notebook Cell 7

('Unnamed: 0_level_0', 'DBSCAN_Cluster')	('CUSTOMER_COUNT', 'Unnamed: 1_level_1')	('CUSTOMER_PERCENT', 'Unnamed: 2_level_1')
-1	8641	96.5500
0	56	0.6300
1	107	1.2000
2	57	0.6400
3	18	0.2000
4	33	0.3700
5	11	0.1200
6	10	0.1100
7	17	0.1900

Table 56 — Generated by Notebook Cell 7

(Unnamed: 0_level_0)	(Borrower Cluster)	(PURCHASES, 'Unnamed: 1_level_0')	(CASH ADVANCE, 'Unnamed: 2_level_0')	(CREDIT LIMIT, 'Unnamed: 3_level_0')	(PREPAYMENT, 'Unnamed: 4_level_0')	(MINIMUM PAYMENT, 'Unnamed: 5_level_0')	(PURCHASE FREQUENCY, 'Unnamed: 6_level_0')	(CASH ADVANCE FREQUENCY, 'Unnamed: 7_level_0')	(PREPAYMENT FREQUENCY, 'Unnamed: 8_level_0')	(CREDIT LIMIT, 'Unnamed: 9_level_1')
0	125.5210	274.9810	0.0000	1823.2140	216.5990	167.8640	0.9610	0.0000	0.0030	11.5540
1	1195.8060	0.0000	135.3940	1518.2240	362.6760	387.6470	0.0000	0.0850	0.0000	11.8600
2	1287.0090	0.0000	190.3840	1490.3510	388.2220	420.5370	0.0000	0.1680	0.0000	11.8950
3	1266.7040	0.0000	146.2120	1383.3330	374.1970	427.3400	0.0000	0.2500	0.0000	12.0000
4	891.8470	64.8730	0.0000	1277.2730	308.6520	345.7750	0.0840	0.0000	0.0000	11.9090
5	1386.2350	0.0000	179.5470	1536.3640	386.3350	457.8590	0.0000	0.3330	0.0000	11.7270
6	0.0000	168.2160	0.0000	1880.0000	113.5060	nan	1.0000	0.0000	0.0000	12.0000
7	31.3470	392.3590	0.0000	2205.8820	383.4020	165.4460	0.9900	0.0000	0.9950	11.6470

Table 57 — Generated by Notebook Cell 7

Unnamed: 0	DBSCAN_Cluster	Customer_Count	Customer_Percentage
0	0	56	0.6300
1	1	107	1.2000
2	2	57	0.6400
3	3	18	0.2000
4	4	33	0.3700
5	5	11	0.1200
6	6	10	0.1100
7	7	17	0.1900

Table 58 — Generated by Notebook Cell 7

('Unnamed: 0_level_0', 'DBSCAN_Cluster')	('ONEOFF_PURCHASES', 'Unnamed: 1_level_1')	('INSTALLMENTS_PURCHASES', 'Unnamed: 2_level_1')	('ONEOFF_PURCHASES_FREQUENCY', 'Unnamed: 3_level_1')	('PURCHASES_INSTALLMENTS_FREQUENCY', 'Unnamed: 4_level_1')
0	0.0000	274.9810	0.0000	0.9380
1	0.0000	0.0000	0.0000	0.0000
2	0.0000	0.0000	0.0000	0.0000
3	0.0000	0.0000	0.0000	0.0000
4	64.8730	0.0000	0.0840	0.0000
5	0.0000	0.0000	0.0000	0.0000
6	0.0000	168.2160	0.0000	1.0000
7	0.0000	392.3590	0.0000	0.9620

Table 59 — Generated by Notebook Cell 7

Unnamed: 0	Metric	Value
0	Total Customers	8950.0000
1	Noise Customers	8641.0000
2	Non-Noise Customers	309.0000
3	Noise Percentage	96.5470

Table 60 — Generated by Notebook Cell 7

Unnamed: 0	Noise_Mean	Overall_Mean	Difference_Percentage
BALANCE	1588.4380	1517.8510	4.6500
PURCHASES	1036.0830	838.8250	23.5160
CASH_ADVANCE	1010.4100	807.1790	25.1780
CREDIT_LIMIT	4598.5040	4490.4280	2.4070
PAYMENTS	1783.3290	1494.0540	19.3620
MINIMUM_PAYMENTS	882.7820	624.0880	41.4520
PURCHASES_FREQUENCY	0.4980	0.4900	1.6070
CASH_ADVANCE_FREQUENCY	0.1370	0.1350	1.2800
PRC_FULL_PAYMENT	0.1570	0.1540	2.2900
TENURE	11.5070	11.5170	-0.0920

Table 61 — Generated by Notebook Cell 7

Unnamed: 0	CHECK	VALUE	EXPECTED	STATUS
0	Customer count preserved	8950.0000	8950	PASS
1	DBSCAN labels assigned	8950.0000	8950	PASS
2	Noise percentage within valid range	96.5500	0% to 100%	PASS
3	At least one non-noise cluster identified	8.0000	>= 1	PASS
4	Non-noise silhouette available	0.5640	Available when >= 2 clusters exist	PASS

Table 62 — Generated by Notebook Cell 7

Unnamed: 0	Metric	Value
0	Selected EPS	0.3000
1	Selected MIN_SAMPLES	10.0000
2	Total Customers	8950.0000
3	Non-Noise Clusters	8.0000
4	Noise Customers	8641.0000
5	Noise Percentage	96.5500
6	Non-Noise Silhouette Score	0.5640

Table 63 — Generated by Notebook Cell 7

Unnamed: 0	CHECK	VALUE	EXPECTED	STATUS	DETAIL
0	Customer count preserved	8950.0000	8950	PASS	nan
1	DBSCAN labels assigned	8950.0000	8950	PASS	nan
2	Noise percentage within valid range	96.5500	0% to 100%	PASS	nan
3	At least one non-noise cluster identified	8.0000	>= 1	PASS	nan
4	Non-noise silhouette available	0.5640	Available when >= 2 clusters exist	PASS	nan
5	Customer count preserved	nan	nan	True	8,950 records
6	DBSCAN labels match customers	nan	nan	True	8,950 labels
7	No missing DBSCAN labels	nan	nan	True	All customers have a DBSCAN label
8	Input index preserved	nan	nan	True	df and df_dbscan indexes match
9	At least two non-noise clusters	nan	nan	True	8 non-noise clusters
10	Non-noise silhouette available	nan	nan	True	0.5638
11	No infinite values in scaled input	nan	nan	True	Scaled matrix contains no infinity
12	No missing values in scaled input	nan	nan	True	Scaled matrix contains no NaN

Table 64 — Generated by Notebook Cell 8

Unnamed: 0	ALGORITHM	CLUSTERS	NOISE_COUNT	NOISE_PERCENT	NON_NOISE_COUNT	COVERAGE_PERCENT	SCENARIOS	DAVIES_BOL	DALINSKI_HAR	SMALLEST_CLUSTER	LARGEST_CLUSTER	SMALLEST_CLUSTER_PERCENT	LARGEST_CLUSTER_PERCENT
0	K-Means	2	0	0.0000	8950	100.0000	0.2260	1.7470	2461.5850	3370	5580	37.6540	62.3460
1	Agglomerative Hierarchical	4	0	0.0000	8950	100.0000	0.3180	0.5010	51.4110	1	8936	0.0110	99.8440
2	DBSCAN	8	8641	96.5480	309	3.4520	0.5640	0.6140	1749.0190	10	107	3.2360	34.6280

Table 65 — Generated by Notebook Cell 8

Unnamed: 0	ALGORITHM	SILHOUETTE	DAVIES_BOULDIN	CALINSKI_HARABASZ	SILHOUETTE_RANK	DAVIES_BOULDIN_RANK	CALINSKI_HARABASZ_RANK
0	K-Means	0.2260	1.7470	2461.5850	3.0000	3.0000	1.0000
1	Agglomerative Hierarchical	0.3180	0.5010	51.4110	2.0000	1.0000	3.0000
2	DBSCAN	0.5640	0.6140	1749.0190	1.0000	2.0000	2.0000

Table 66 — Generated by Notebook Cell 8

Unnamed: 0	ALGORITHM	CLUSTERS	NON_NOISE_CUSTOMERS	AVERAGE_PERCENTAGE	NOISE_COUNT	NOISE_PERCENT	SMALLEST_CLUSTER	LARGEST_CLUSTER	SMALLEST_CLUSTER_PERCENT	LARGEST_CLUSTER_PERCENT
0	K-Means	2	8950	100.0000	0	0.0000	3370	5580	37.6540	62.3460
1	Agglomerative Hierarchical	2	8950	100.0000	0	0.0000	1	8936	0.0110	99.8440
2	DBSCAN	8	309	3.4530	8641	96.5470	10	107	3.2360	34.6280

Table 67 — Generated by Notebook Cell 8

Unnamed: 0	KMeans_Cluster	CUSTOMER_COUNT	CUSTOMER_PERCENT
0	0	5580	62.3460
1	1	3370	37.6540

Table 68 — Generated by Notebook Cell 8

Unnamed: 0	Agg_Cluster	CUSTOMER_COUNT	CUSTOMER_PERCENT
0	0	8936	99.8440
1	1	12	0.1340
2	2	1	0.0110
3	3	1	0.0110

Table 69 — Generated by Notebook Cell 8

Unnamed: 0	DBSCAN_Cluster	CUSTOMER_COUNT	CUSTOMER_PERCENT
0	-1	8641	96.5470
1	0	56	0.6260
2	1	107	1.1960
3	2	57	0.6370
4	3	18	0.2010
5	4	33	0.3690
6	5	11	0.1230
7	6	10	0.1120
8	7	17	0.1900

Table 70 — Generated by Notebook Cell 8

('Agg_Cluster', 'KMeans_Cluster')	('0', 'Unnamed: 1_level_1')	('1', 'Unnamed: 2_level_1')	('2', 'Unnamed: 3_level_1')	('3', 'Unnamed: 4_level_1')
0	5566	12	1	1
1	3370	0	0	0

Table 71 — Generated by Notebook Cell 8

Unnamed: 0	KMEANS_CLUSTER	DOMINANT_AGG_CLUSTER	DOMINANT_SHARE_PERCENT
0	0	0	99.7490
1	1	0	100.0000

Table 72 — Generated by Notebook Cell 8

Unnamed: 0	LOW_DENSITY_MEAN	OVERALL_MEAN	DIFFERENCE_PERCENT
BALANCE	1540.1470	1517.8510	1.4690
PURCHASES	865.8250	838.8250	3.2190
CASH_ADVANCE	832.5780	807.1790	3.1470
CREDIT_LIMIT	4594.3270	4490.4280	2.3140
PAYMENTS	1535.6900	1494.0540	2.7870
MINIMUM_PAYMENTS	634.2620	624.0880	1.6300
PURCHASES_FREQUENCY	0.4980	0.4900	1.6070
CASH_ADVANCE_FREQUENCY	0.1370	0.1350	1.2800
PRC_FULL_PAYMENT	0.1570	0.1540	2.2900
TENURE	11.5070	11.5170	-0.0920

Table 73 — Generated by Notebook Cell 8

Unnamed: 0	ALGORITHM	STRENGTH	LIMITATION	OBSERVED_RESULT	BUSINESS_USE
0	K-Means	Simple, scalable, interpretable and assigns	Requires a predefined number of clusters	Produce two balanced customer groups	Whole customer segmentation and behavioural ta...
1	Agglomerative Hierarchical	Provides hierarchical structure and an inde	The tested four-cluster solution is extremely	Four clusters are produced, but 99.844% o	Structural validation and examination of custo...
2	DBSCAN	Can identify dense groups and separate low	Sensitive to eps and min_samples and can	Only non-noise clusters are identified, but	Density-based behavioural investigation and sp...

Table 74 — Generated by Notebook Cell 8

(Unnamed: 0_level_0)	BALANCE (Unit)	PURCHASES (Unit)	CASH ADVANCE (Unit)	CREDIT LIMIT (Unit)	PAYMENTS (Unit)	MINIMUM PAYMENT (Unit)	PURCHASE FREQUENCY (SM)	CASH ADVANCE FREQUENCY (SM)	PAYMENT FREQUENCY (SM)	CREDIT LIMIT (Unit)
0	870.0050	1152.3300	119.4140	4304.5710	1340.2930	447.1450	0.6770	0.0270	0.2250	11.6070
1	2590.5450	319.7270	1945.9710	4798.1670	1748.6510	917.0670	0.1810	0.3140	0.0360	11.3680

Table 75 — Generated by Notebook Cell 8

Unnamed: 0	KMEANS_CLUSTER	CUSTOMER_COUNT	CUSTOMER_PERCENT	BEHAVIOUR
0	0	5580	62.3460	High purchase activity; Low cash-advance usage...
1	1	3370	37.6540	Low purchase activity; High cash-advance usage...

Table 76 — Generated by Notebook Cell 8

Unnamed: 0	SEGMENT	CUSTOMERS	CUSTOMER_SHARE_%	RELATIVELY_HIGHER_BEHAVIOUR	RELATIVELY_LOWER_BEHAVIOUR
0	0	5580	62.3500	PRC_FULL_PAYMENT, PURCHASES_FREQUENCY, PURCHASE_CASES	ADVANCE_FREQUENCY, BALANCE_FREQUENCY, PAYMENT_FREQUENCY
1	1	3370	37.6500	CASH_ADVANCE, CASH_ADVANCE_FREQUENCY, BALANCE_FREQUENCY, ADVANCE_FREQUENCY	PURCHASE_CASES, PURCHASE_FREQUENCY, PURCHASE_FULL_PAYMENT

Table 77 — Generated by Notebook Cell 8

Unnamed: 0	SEGMENT	POTENTIAL_BUSINESS_ACTIONS
0	0	Consider purchase-reward or loyalty campaigns....
1	1	Review cash-advance usage and consider financi...

Table 78 — Generated by Notebook Cell 8

Unnamed: 0	CHECK	VALUE	EXPECTED	STATUS
0	Customer count preserved	8950.0000	8950	PASS
1	K-Means labels match customers	8950.0000	8950	PASS
2	Agglomerative labels match customers	8950.0000	8950	PASS
3	DBSCAN labels match customers	8950.0000	8950	PASS
4	K-Means contains multiple clusters	2.0000	>= 2	PASS
5	Agglomerative contains multiple clusters	4.0000	>= 2	PASS
6	DBSCAN contains multiple non-noise clusters	8.0000	>= 2	PASS
7	K-Means business profile created	2.0000	2	PASS
8	Agglomerative cluster balance reviewed	99.8440	< 99% preferred	WARNING
9	DBSCAN coverage reviewed	3.4530	Context required	WARNING

Table 79 — Generated by Notebook Cell 9

Unnamed: 0	FEATURE_ORDER	FEATURE_NAME
0	1	BALANCE
1	2	BALANCE_FREQUENCY
2	3	PURCHASES
3	4	PURCHASES_FREQUENCY
4	5	PURCHASES_TRX
5	6	CASH_ADVANCE
6	7	CASH_ADVANCE_FREQUENCY
7	8	CASH_ADVANCE_TRX
8	9	CREDIT_LIMIT
9	10	PAYMENTS
10	11	MINIMUM_PAYMENTS
11	12	PRC_FULL_PAYMENT
12	13	PURCHASE_TO_CREDIT_RATIO
13	14	ONEOFF_PURCHASE_SHARE
14	15	INSTALLMENT_PURCHASE_SHARE

Table 80 — Generated by Notebook Cell 9

Unnamed: 0	SETTING	VALUE
0	Infinite values	Replace ±inf with NaN
1	Missing-value treatment	Median imputation
2	Outlier method	3.0 × IQR capping
3	Outlier multiplier	3.000
4	Skewness transformation	log1p on: PURCHASES_TRX, CASH_ADVANCE_TRX
5	Scaling method	StandardScaler
6	Random state	42
7	K-Means clusters	2

Table 81 — Generated by Notebook Cell 9

Unnamed: 0	BALANCE	BALANCE	PURCHASES	ONEOFF_PURCHASES	UNRECORDED_PURCHASES	REWARDS	PURCHASES	REWARDS	REWARDS	REWARDS	REWARDS	REWARDS	REWARDS	REWARDS	REWARDS	REWARDS	REWARDS	REWARDS
0	40.9010	0.8180	95.4000	0.0000	95.4000	0.0000	0.1670	0.0000	0.0830	0.0000	0	2	1000.0000	201.8020	139.5100	0.0000	12	0
1	3202.4670	0.9090	0.0000	0.0000	0.0000	6442.9450	0.0000	0.0000	0.0000	0.2500	4	0	7000.0000	4103.0330	1072.3400	0.2220	12	1
2	2495.1490	1.0000	773.1700	773.1700	0.0000	0.0000	1.0000	1.0000	0.0000	0.0000	0	12	7500.0000	622.0670	627.2850	0.0000	12	0
3	1666.6710	0.6360	1499.0000	1499.0000	0.0000	205.7880	0.0830	0.0830	0.0000	0.0830	1	1	7500.0000	0.0000	nan	0.0000	12	0
4	817.7140	1.0000	16.0000	16.0000	0.0000	0.0000	0.0830	0.0830	0.0000	0.0000	0	1	1200.0000	678.3350	244.7910	0.0000	12	0
5	1809.8290	1.0000	1333.2800	0.0000	1333.2800	0.0000	0.6670	0.0000	0.5830	0.0000	0	8	1800.0000	1400.0580	2407.2460	0.0000	12	0
6	627.2610	1.0000	7091.0100	6402.6300	688.3800	0.0000	1.0000	1.0000	1.0000	0.0000	0	64	13,500.00	6354.3140	198.0660	1.0000	12	0
7	1823.6530	1.0000	436.2000	0.0000	436.2000	0.0000	1.0000	0.0000	1.0000	0.0000	0	12	2300.0000	679.0650	532.0340	0.0000	12	0
8	1014.9260	1.0000	861.4900	661.4900	200.0000	0.0000	0.3330	0.0830	0.2500	0.0000	0	5	7000.0000	688.2790	311.9630	0.0000	12	0
9	152.2260	0.5450	1281.6000	1281.6000	0.0000	0.0000	0.1670	0.1670	0.0000	0.0000	0	3	11,000.00	1164.7710	100.3020	0.0000	12	0

Table 82 — Generated by Notebook Cell 9

Unnamed: 0	KMEANS_CLUSTER	CUSTOMER_COUNT	CUSTOMER_PERCENT
0	0	5574	62.2790
1	1	3376	37.7210

Table 83 — Generated by Notebook Cell 9

Unnamed: 0	METADATA	VALUE
0	project_name	Credit Card Customer Segmentation
1	model_type	K-Means
2	number_of_clusters	2
3	random_state	42
4	max_iterations	500
5	n_init	20
6	number_of_customers	8950
7	number_of_clustering_features	15
8	clustering_features	[BALANCE, BALANCE_FREQUENCY, PURCHASES, PURCHA...
9	scaling_method	StandardScaler
10	outlier_method	IQR capping
11	outlier_iqr_multiplier	3.000
12	skewness_threshold	2.000
13	log_transformed_features	[PURCHASES_TRX, CASH_ADVANCE_TRX]
14	deployment_test_match_percentage	99.866
15	training_silhouette	0.226
16	training_inertia	105285.958